

RBM7 deficiency promotes breast cancer metastasis by coordinating MFGE8 splicing switch and NF- κ B pathway

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Abstract

Aberrant alternative splicing is well-known to be closely associated with tumorigenesis of various cancers. However, the intricate mechanisms underlying breast cancer metastasis driven by deregulated splicing events remain largely unexplored. Here, we unveiled RBM7 as a novel regulator of alternative splicing that plays a crucial role in counteracting the metastatic potential of breast cancer. Through bioinformatics analysis and IHC staining validation, we revealed that RNA binding motif protein 7 (RBM7) is decreased in lymph node and distant organ metastases of breast cancer as compared to primary lesions. Furthermore, we found that low expression of RBM7 is correlated with the reduced disease-free survival of breast cancer patients. Breast cancer cells with RBM7 depletion exhibited an increased potential for lung metastasis compared to scramble control cells. The absence of RBM7 stimulated breast cancer cell migration, invasion, and angiogenesis. Mechanistically, RBM7 controlled the splicing switch of MFGE8, favoring the production of the predominant isoform of MFGE8, MFGE8-L. This resulted in the attenuation of STAT1 phosphorylation and alterations in cell adhesion molecules. MFGE8-L exerted an inhibitory effect on the migratory and invasive capability of breast cancer cells, while the truncated isoform MFGE8-S, which lack the second F5/8 type C domain had the opposite effect. Particularly, the ectopic expression of MFGE8-L significantly reversed the pro-invasion effect of RBM7 silencing, but did not contribute to the promotion of angiogenesis-related secreted proteins. In RBM7-depleted cells, a gene set enrichment analysis revealed a significant amplification of the NF- κ B cascade. Concordantly, RBM7 negatively regulates p65 phosphorylation. Furthermore, an NF- κ B inhibitor could obstruct the increase in HUVEC tube formation caused by RBM7 silencing. Clinically, we noticed a positive correlation between RBM7 expression and MFGE8 exon7 inclusion in breast cancer tissues. Therefore, our study not only offer mechanistic insights into how abnormal splicing contributes to the aggressiveness of breast cancer, but also provide a new approach for molecular-targeted therapy in combating breast cancer.

eLife assessment

This study presents a rather **valuable** finding on the RBM7 function in splicing regulation and uncharacterized role of MFGE8 splicing alteration in breast cancer metastasis. The evidence supporting the claims of the authors is **solid**. The work will be of broad interest to clinicians, medical researchers and scientists working on breast cancer.

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Introduction

Alternative splicing (AS) emerges as a substantial contributor to the production of a myriad of transcripts from limited genes, which enriched protein variation and phenotypic diversity in multicellular organisms. Pre-mRNA splicing reactions require a step-wise assembly of spliceosome, a megadalton complex that recognize consensus regulatory sequences, including 5' and 3' splice sites marking exon-intron boundary, and the branch point site (BPS), to accomplish exon definition (Shi, 2017 [↗](#)) (Wilkinson, Charenton et al., 2020). While a splice site is suboptimal, AS emerges under control of inherent sequence acting in cis, namely exonic or intronic splicing enhancer or silence, and the cognate trans-acting splicing factors, which can strengthen or weaken the recognition of splice sites by spliceosome.

It is well documented that AS dysregulation leads to multiple human diseases, including cancer (Montes, Sanford et al., 2019) (Singh & Cooper, 2012 [↗](#)). Tumorigenicity has been linked with cancer-specific splicing switch arising from either cis-elements mutations or deregulated splicing regulatory factors. In some cases, splicing factors, particularly spliceosomal components SF3B1 and U2AF1, are frequently mutated across diverse cancer types (Baralle, 2005 [↗](#)) (Harbour, Roberson et al., 2013) (Smith, Choudhary et al., 2019) (Cheruiyot, Li et al., 2021 [↗](#)). For instance, recurrent SF3B1 mutations led to retention of intronic region in BRD9 pre-mRNA, conferring loss of core subunit function to the ncBAF chromatin remodeling complex and metastatic advantage to melanoma cells (Inoue, Chew et al., 2019). Another layer of regulation involves the abnormal expression of splicing factors derived from altered regulatory pathway. As an archetypal example, a set of members in SR protein family such as SRSF1 and SRSF3 are upregulated in breast tumors, lung tumors, glioma and exhibit oncogenic ability via altering AS products (for example, Bcl-x, MYO1B, NUMB and p53) (Anczuków, Akerman et al., 2015) (Zhou, Wang et al., 2019) (Tang, Horikawa et al., 2012) (Zhou, Gong et al., 2020). Otherwise, many splicing factors such as RBM (RNA binding motif) proteins can either functions as oncoproteins or tumor suppressors in cancer-type dependent manner (Li, Guo et al., 2021), which may be attributed to tissue-specific AS patterns.

Breast cancer is the most common diagnosed malignancy and remain the second leading cause of cancer mortality (over 41,000 deaths annually in the USA) among females worldwide (Siegel, Miller et al., 2019). Metastasis to vital organs accounts for approximately 90% cancer-related death. The 5-year overall survival rate of patients with metastatic breast cancer is drastically decreased to about 25%, as compared to the primary cancer patients' prognosis with a survival rate greater than 80% (Rabbani & Mazar, 2007 [↗](#)) (Valastyan & Weinberg, 2011). Despite advances in early diagnosis and comprehensive interventions for breast cancer treatment, limited effective strategy against metastasis has been an outstanding challenge for improving prognostication of patients. This highlights the urgent need for deepened understanding of mechanisms underlying development of metastasis. Tumor metastasis is multi-step process involving a sequential cascade

of events, such as epithelial-mesenchymal transition (EMT), local invasion from primary tumor site, lymphogenic or hematogenic dissemination, cancer cells seeding and metastatic lesions formation (Liang, Zhang et al., 2020). AS dysregulation has been implicated in multistage process of metastasis. For example, the switch of CD44 or FGFR2 splicing isoforms can lead to EMT phenotype (Chen, Zhao et al., 2018) (Warzecha, Sato et al., 2009). VEGF-A or XBP1 splicing transition is tightly linked to angiogenesis in cancer (Bates, 2008) (Zeng, Xiao et al., 2013). On account of the phenotypic plasticity in metastatic cells conferred by alternative splicing, further investigation in potential role of critical splicing regulatory factors and the associated splicing events in breast cancer metastasis is warranted.

RNA-binding proteins (RBPs) are a large class of conserved proteins that participate in cancer development through precisely control RNA metabolism, including RNA splicing. RBM7, a member of RBP family, is commonly known to exert RNA surveillance activity as a component of the human nuclear exosome target (NEXT) complex (Meola, Domanski et al., 2016). Aberrant expression of RBM7 is not only associated with lung fibrosis onset and motor neuronopathy, but also impacts cancer cell proliferation (Fukushima, Satoh et al., 2020) (Xi, Zhang et al., 2020). Previous findings have established a direct connection of RBM7 with spliceosome factors SF3B2 and SF3B4 (Falk, Finogenova et al., 2016), yet whether RBM7 is involved in the regulation of AS remain elusive. In this study, we report that RBM7 is frequently reduced in a subset of breast cancer metastases from lymph nodes lesions and distant organ as compared to primary tumors and its low expression predicts dismal survival outcomes of patients. Breast cancer cells with loss of RBM7 gain enhanced aggressive capability and metastatic potential. Mechanistically, RBM7 ablation enhanced the splicing switch of MFGE8 pre-mRNA from canonical isoform MFGE8-L to a truncated isoform MFGE8-S. Ectopic expression of MFGE8-L caused marked suppression of migratory and invasive ability of breast cancer cells, whereas MFGE8-S showed opposite function/disabled such function. At another layer, RBM7 discordantly regulated the phosphorylation of p65. In concert, NF- κ B inhibitor abolished pro-angiogenesis effect of RBM7 ablation in breast cancer. Above all, our findings uncovered a previously unidentified role of RBM7 in breast cancer metastasis via coupling MFGE8 splicing switch and NF- κ B signaling activation, suggesting a new therapeutic target and prognostic predictor for breast cancer.

Results

RBM7 is reduced in metastatic lesions of breast cancer and positively correlated with patients' prognosis

To determine the clinical relevance of RBM7 in breast cancer, we assessed the correlation between RBM7 and important clinical outcomes. Analysis of TCGA, GEO and the METABRIC breast cancer datasets showed that RBM7 expression positively correlates with overall survival of breast cancer patients (**Figure 1A** [↗](#) and Supplementary Figure 1A-1B). RBM7 low expression predicts the reduced disease-free survival in patients with breast cancer (**Figure 1B** [↗](#) and Supplementary Figure 1C). Congruently, we found RBM7 is decreased in breast cancer compared to normal tissues (**Figure 1C** [↗](#) and Supplementary Figure 1D). Further dissection in a cohort of breast cancer patients from the cBioPortal unveiled cases with lymph node metastasis have lower expression of RBM7 as compared to cases without metastasis (**Figure 1D** [↗](#)). We next performed immunohistochemistry (IHC) detection of RBM7 in breast cancer tissue microarrays. The results showed that patients bearing clinical stage III tumors have significantly decreased RBM7 expression than patients bearing I and II clinical stage tumors (**Figure 1E** [↗](#)). In tissue chip containing triple negative breast cancer, the more aggressive and mesenchymal breast cancer subtype, RBM7 exhibits reduced protein levels in tumors as compared to paracancer tissues (**Figure 1F** [↗](#)). In addition, we collect specimen consisted of tumor tissues from 21 breast primary cancer, in 9 cases, from metastatic loci and took advantage of IHC analysis to determine the RBM7 expression. Most (25% strong positive; 67% moderate positive) of primary tumor samples displayed

a positive IHC staining for RBM7, whereas most of lung/liver metastatic lesions showed weak or negative RBM7 expression (**Figure 1G** [↗](#)). We also obtained paired primary breast cancer and lymph node metastatic loci and found, when compared to the corresponding primary tumor, the expression of RBM7 is significantly decreased in lymph node metastases (**Figure 1H** [↗](#)-**1I** [↗](#)). These results associated RBM7 with breast cancer metastasis, prompting us to further investigate the functional role of RBM7 in metastatic processes of breast cancer.

RBM7 silencing provokes metastatic potential of breast cancer

In light with the above findings, we generated RBM7-silenced breast cancer cells using lentiviral expression of short hairpin RNA (shRNA) followed by profiling high-throughput transcriptome by RNA sequencing. Strikingly, Gene Ontology and pathway analysis displayed a wide range of differentially expressed genes were enriched in extracellular matrix organization, cell migration and angiogenesis (**Figure 2A** [↗](#)-**2B** [↗](#)), which have been characterized as crucial processes in tumor metastasis. We then conducted qRT-PCR analysis of these metastasis-associated genes in RBM7-depleted breast cancer cells and verified that they were up-regulated upon RBM7 knockdown shown as heatmap representation (**Figure 2C** [↗](#)). The lung metastasis is one of the most prevalent distant metastases that breast cancer is prone to and associated with poor prognosis of patients (Medeiros & Allan, 2019 [↗](#)). Thus, we performed tail-vein transfer model to evaluate the potential impact of RBM7 on pulmonary metastasis of breast cancer cells. 6×10^5 4T1 cells, the mouse breast cancer cell line simulating stage IV human breast tumor, with stable RBM7 silence or control vector were implanted into BALB/c mice via tail vein. RBM7-silenced 4T1 cells were able to develop more spontaneous lung metastases than control cells as judged by pulmonary nodules number and haematoxylin-eosin (HE) staining of lung tissues (**Figure 2D** [↗](#)-**2F** [↗](#) and Supplementary Figure 2A). Cancer metastasis is comprised of a complex cascade of events, such as migration, invasion and angiogenesis, thus we next to determine which process is mediated by RBM7-loss in facilitating breast cancer metastasis. We utilized transwell chamber with or without matrix coat and wound healing assays to examine the potential function of RBM7 in migration and invasion capabilities and found the depletion of RBM7 led to a visible increase in mobility and invasiveness of both the triple-negative breast cancer (TNBC) cell line MDA-MB-231, estrogen receptor positive breast cancer cell line MCF7 and 4T1 cells (**Figure 2G** [↗](#)-**2H** [↗](#) and Supplementary Figure 2B-2E). Conversely, RBM7 overexpression notably repressed MDA-MB-231 and MCF7 cells to migrate or invade across transwell inserts (**Figure 2I** [↗](#)-**2J** [↗](#)). Similar results were obtained in two additional TNBC cell lines BT549 and HCC1937 (Supplementary Figure 2F-2G), which proved that RBM7 possessed the capability to suppress the migration and invasion of breast cancer cells. To examine the impact of RBM7 on angiogenesis in breast cancer, we cultured human umbilical vein endothelial cells (HUVEC) with condition medium obtained from culture supernatants of RBM7-depleted or scramble shRNA transduced cells for tube formation analysis. Compared to the scramble control, HUVEC incubated with culture medium from RBM7-knockdown breast cancer cells exhibited accelerated tube formation length and branch junction (**Figure 2K** [↗](#)), suggesting the inhibitory effect of RBM7 on breast cancer angiogenesis. Taken together, the above data demonstrated RBM7 as tumor suppressor by regulating breast cancer migration, invasion and angiogenesis.

RBM7 exhibits splicing regulatory function in breast cancer

In previous studies, RBM7 is commonly characterized as an RNA-binding constituent of trimeric NEXT complex, which directs the RNA exosome to non-coding transcriptome for degradation (Lubas, Andersen et al., 2015). However, the mechanistical role of RBM7 in cancer remain largely unknown. Intriguingly, we observed that, attributed to the ablation of RBM7, almost 551 splicing events occurred significant decrease in PSI (percent spliced in), whereas 289 splicing events endured increased PSI (**Figure 3A** [↗](#)). The changed events were divided among various types of AS (alternative splicing), including SE (skipping exon), IR (intron retention), A5E (alternative 5' splice site exon), A3E (alternative 3' splice site exon), and MXE (mutually exclusive exons), of note, SE

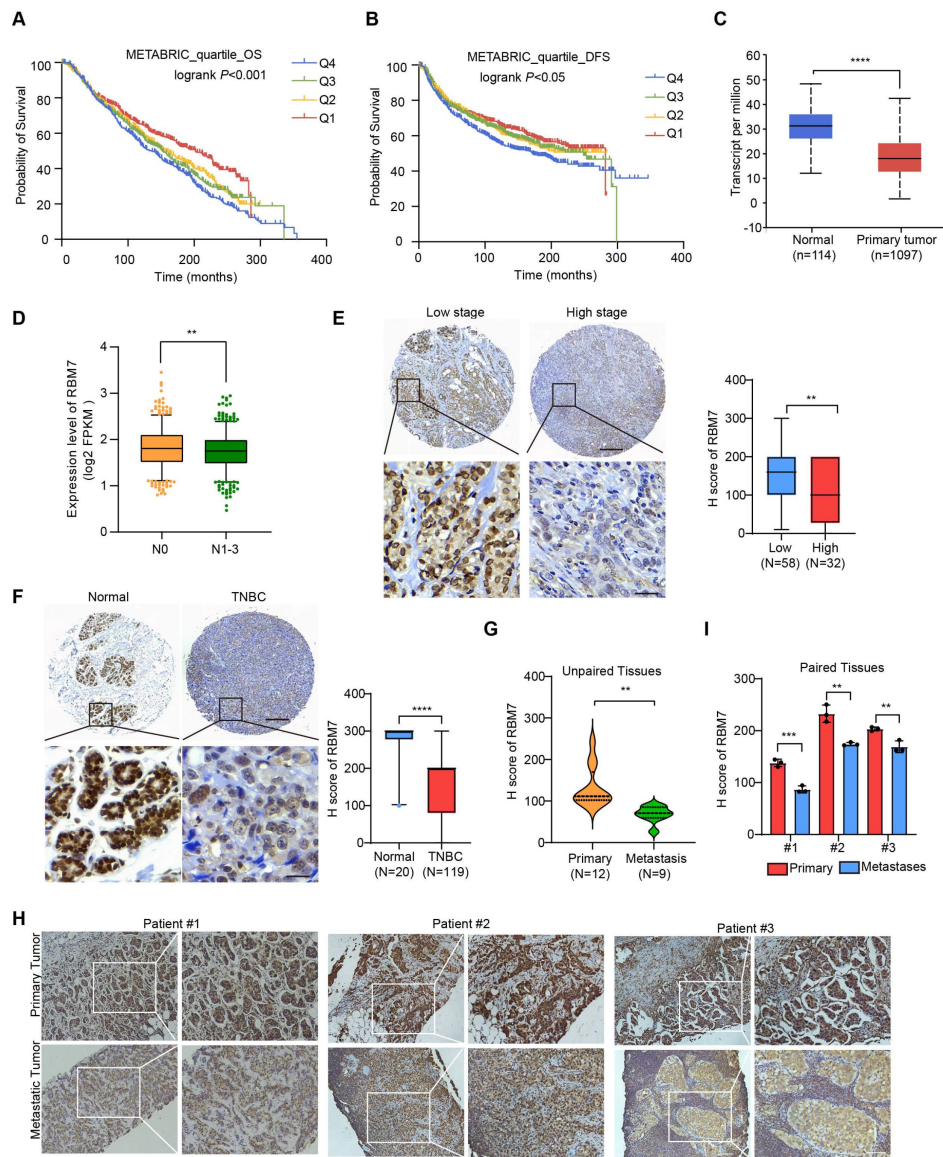


Figure 1.

Decreased RBM7 expression was associated with poor prognosis of breast cancer.

(A-B) The correlation between RBM7 mRNA expression and overall survival (OS) or disease-free-survival (DFS) of breast cancer patients ($n=1980$) was analyzed based on the METABRIC dataset. The samples were divided into four equal parts, containing lower quartile Q1, median quartile Q2, upper quartile Q3 and higher quartile Q4 according to the expression of RBM7. Data are presented as means $\pm$ SD and P values were obtained by Mantel-Cox log-rank test. (C) The expression of RBM7 in primary breast carcinoma ($n=1097$) compared to normal tissues ($n=114$) was analyzed through UALCAN dataset. (D) Analysis of RBM7 expression in breast cancer lymph node metastases in comparison with breast cancer tissues with no lymph node metastasis based on TCGA dataset. BRCA samples were classified into N0 (No regional lymph node involvement) ($n=515$), and metastases in lymph node (N1-N3) ($n=565$). (E) Representative IHC images of RBM7 staining for patients at high ($n=32$) or low ($n=58$) clinical stages on tissue microarray of breast cancer specimens. Scale bars = 300 μ m (top) or 30 μ m (bottom). (F) Representative images of the IHC staining of RBM7 in a tissue microarray containing triple-negative breast cancer ($n=119$) and para-carcinoma tissues ($n=20$). Scale bars = 300 μ m (top) and 30 μ m (bottom). (G) Quantitative analysis of RBM7 expression according to IHC staining scores in primary breast cancer ($n=12$) and distant metastases (breast cancer lung metastases, $n=4$; breast cancer liver metastases, $n=5$). (H-I) Representative IHC images and quantitative analyses for RBM7 staining in 3 paired primary breast cancer and lymphatic metastases are shown. Scale bar = 100 μ m. Data are presented as means $\pm$ SD and P values were obtained by unpaired Student t test (C-I). $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$.

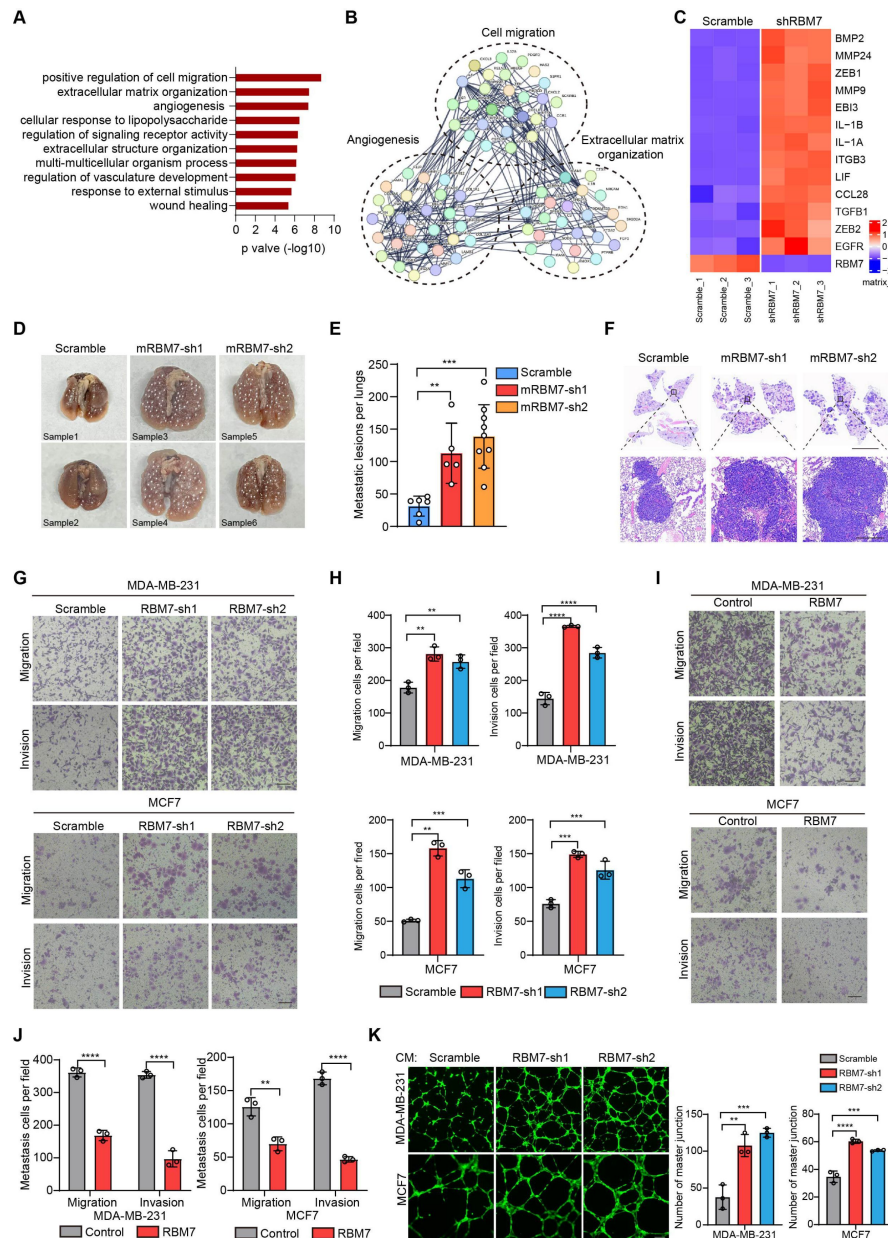


Figure 2.

RBM7 negatively regulated breast cancer metastatic potential. **(A)** Gene Ontology analysis showed the significantly affected biological process upon RBM7 knockdown in MDA-MB-231 cells. **(B)** Functional association network of RBM7-regulated targets. The genes were analyzed using the STRING database, and subgroups are marked according to their function. **(C)** A heatmap showing the qRT-PCR analysis of differentially expressed genes upon RBM7 knockdown in breast cancer cells from three biological replicates. **(D-F)** 4T1 cells without or with RBM7 knockdown were injected into tail vein of immunodeficient BALB/c mice to establish a lung metastasis model. Spontaneous pulmonary metastases were assessed after about 3 weeks, the white arrow indicates lung lesions macroscopically **(D)**. H&E stained lung sections were quantified for the number of spontaneous metastatic lesions from BALB/c mice (n=6, 5 and 9). Data are presented as mean $\pm$ SD and P values were determined by one-way ANOVA with Dunnett's multiple comparisons test **(E)**. Representative images of H&E stained lung sections are shown. Top panel scale bars indicate 7 mm; bottom panel scale bars indicate 300 μ m **(F)**. **(G-J)** The metastatic ability of breast cancer cells with RBM7 depletion **(G)** or ectopic expression of RBM7 **(I)** were evaluated by transwell assay. Scale bars: 100 μ m. Data are mean $\pm$ SD from three random fields. P values were determined by one-way ANOVA with Dunnett's multiple comparisons test **(H)** or unpaired Student's T test **(J)**. **(K)** Representative images of the tube formation of HUVEC cells treated with conditional medium from RBM7-KD cells or control cells for 12h. Scale bars: 100 μ m. Quantification of number of junctions formed by HUVEC was calculated by Image J software. ****** $P < 0.01$, ******* $P < 0.001$, ******** $P < 0.0001$.

events were the most prominently regulated (**Figure 3B**). A majority of AS events in particular of SE and ASE category were in decrease of PSI upon RBM7 depletion (**Figure 3C-3D**), indicating that the presence of RBM7 may promote the usage of alternative splice sites of SEs or ASEs. Subsequently, we extracted the candidate binding motifs which reside within or near cassette exons or ASE and found UUUCUU motif, similar to RBM7 binding site (polyU) were enriched in RBM7 positively regulated AS events to a greater extent than control exons unimpacted by RBM7 (**Figure 3E**), raising the possibility that RBM7 may serve as a splicing activator through directly tethering cis-element in pre-mRNA. This is coherent with the nuclear localization of RBM7 in breast cancer cells (Supplementary Figure 3A). In order to explore the splicing regulation activity of RBM7 in distinct pre-mRNA context, we validated a series of RBM7-significantly-changed splicing events across different AS types according to the RNA-seq results. The RT-PCR analysis demonstrated that the PSI of selected gene related to cancer progression was either increased or decreased endogenously upon RBM7 depletion (**Figure 3F** and Supplementary Figure 3B). Considering that the AS is a sophisticated multi-step reaction that is regulated by both pre-mRNA-interacting splicing factor and other assistant factors in a context-dependent manner (Wright, Smith et al., 2022), we doubted the divergent effect of RBM7 on distinct splicing targets was due to a more complicated mechanism. Gene ontology (GO) enrichment analysis identified a majority of genes undergoing AS alterations were involved in cytoskeleton assembly and organization (**Figure 3G**). While globally analyzing the network enrichment of RBM7-regulated downstream splicing events, we observed its targets are functionally connected to well-linked network containing gene subgroups (**Figure 3H**), which are associated with biological processes in tumor cell metastasis, again suggesting the cancer-related function of RBM7.

RBM7 depletion switches splicing outcomes of MFGE8 by decreasing MFGE8-L isoform

To ascertain the functional targets that explain the inhibitory role of RBM7 in breast cancer metastasis, we analyzed RNA immunoprecipitation sequencing data for RBM7 and whole-transcriptome sequencing data for breast cancer cells expressing two shRNAs targeting RBM7. Venn diagram of 8 hub genes obtained by overlapping three gene sets pointed forward a hotspot gene, namely MFGE8 (**Figure 4A**). RNA-seq data showed that the depletion of RBM7 resulted in a significant alteration in ratio of MFGE8 two isoforms, of which the canonical isoform (MFGE8-L) has been demonstrated to be correlated with tumor formation and aggressiveness of breast cancer (Yang, Hayashida et al., 2011). Importantly, the exon 7 located in F5/8 type C 2 domain, which is obligated to bind to phosphatidylserine, was lacked in the short isoform of MFGE8 (MFGE8-S) (**Figure 4B**). Further RT-PCR identification verified RBM7 ablation indeed potentiated the production of MFGE8-S in various TNBC cell lines (**Figure 4C**). Similar results were obtained from MCF7 cells, suggesting that RBM7 regulated the AS of MFGE8 independently of the ER status. In separate experiments, we transfected breast cancer cells with an exogenous MFGE8 splicing reporter containing exon 6-8 and 600bp of flanking intron sequences to determine the splicing regulatory function of RBM7. Accordingly, the depletion of RBM7 still induced the splicing shift of the minigene reporter by elevating MFGE8-S variant (**Figure 4D**). To test whether such splicing regulatory effect was directly or indirectly, we analyzed the sequence of alternative exon 7 and the motifs nearby its 5' or 3' splice sites, and found two poly(U) motifs are positioned in proximal to the pseudo 3' splice site. Subsequent RT-PCR assay for the precipitation in RIP assays confirmed RBM7 could bind to the upstream sequence containing UUUCUU motifs adjacent to 3' splice site of MFGE8 cassette exon, but not another region nearby it (**Figure 4E**). To pinpoint the location for the potential cis-element for AS regulation by RBM7, we designed antisense oligonucleotides (ASOs), which have been chemically modified to avoid endogenous nuclease degradation without recruiting RNase H, to block the UUUCUU residues in proximity to intron6/exon7 junction. As shown in **Figure 4F**, when compared to control ASO, treatment with ASOs targeting the potential binding site (UUUCUU) of RBM7, contributed to the exclusion of exon7 of MFGE8. In separate experiments, the depletion of RBM7 induced the splicing shift of the minigene reporter by

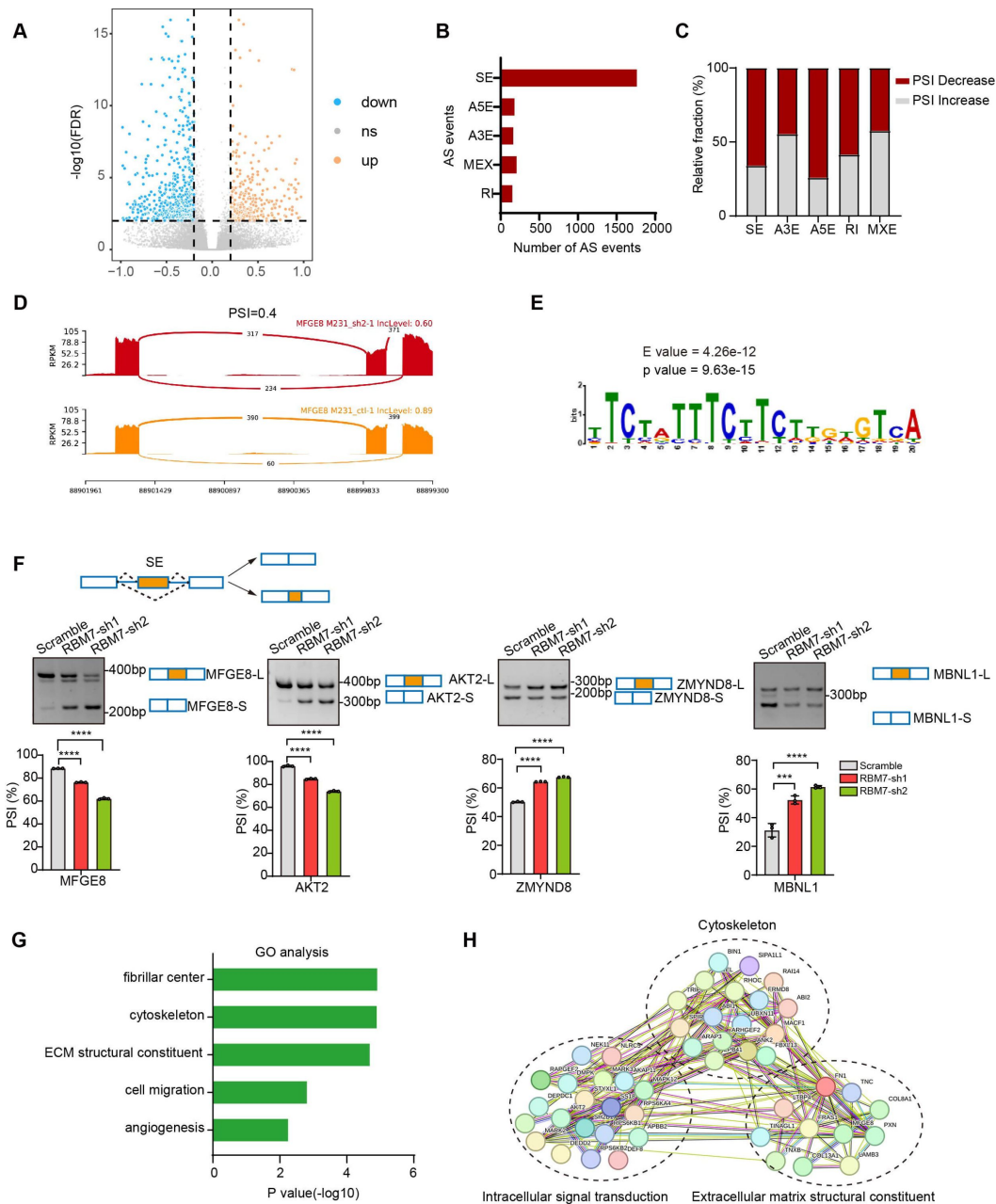


Figure 3.

Global identification of alternative splicing events regulated by RBM7.

(A-E) RNA-seq analysis was performed on RBM7-knockdown MDA-MB-231 cells or control cells, and the changes in splicing events were analyzed. (A) Volcano plot illustrating up-regulated/down-regulated alternative splicing events upon RBM7 depletion. (B) Comparison of the differential splicing events in five types of AS events affected by RBM7 depletion. (C) The relative fraction of splicing events undergoing percent spliced in (PSI) increase or decrease induced by RBM7 depletion. (D) Alternative splicing of MFGE8 was chosen to represent a decrease of PSI, and numbers of exon junction reads were indicated. (E) The hit top enriched motifs identified in the differential spliced genes regulated by RBM7. (F) RBM7-regulated exon skipping events were identified by semiquantitative RT-PCR using RBM7-depleted or control MDA-MB-231 cells. The mean $\pm$ SD of PSI from 3 independent experiments was plotted, with *P* values determined by one-way ANOVA with Dunnett's multiple comparisons test. (G) Gene Ontology analysis showed the significantly enriched biological processes of changed splicing events affected upon RBM7 knockdown in MDA-MB-231 cells. (H) KEGG pathway enrichment analysis of functional association network of the RBM7-controlled AS targets.

elevating MFGE8-S variant. While the binding motif was depleted or mutated, RBM7 failed to affect the splicing outcomes of MFGE8 (Supplementary Figure SC). Due to its close proximity to 3' splice site, UUUCUU residues bound by RBM7 might participated in spliceosome assembly on the upstream 3' splice site of exon7, which may explain why disruption of the motif led to almost complete exon7 skipping. Our data suggest that RBM7 regulates the exon skipping of MFGE8 by directly binding to UUUCUU located at six nucleotides upstream of the 3' splice-site of exon7.

The short variant of MFGE8 abolishes the inhibitory function of its classical isoform on breast cancer cell migration and invasion

MFGE8 encodes a secreted protein, Milk fat globule-EGF factor 8 (MFG-E8), which participates in a variety of cellular processes, including mammary gland branching morphogenesis, angiogenesis and mucosal healing, etc (Raymond, Ensslin et al., 2009). We applied MFGE8-L stably overexpressing- or depleted-breast cancer cells to transwell assays and observed the cellular migratory and invasive capabilities was visibly suppressed by MFGE8-L overexpression (Supplementary Figure 4A-4B). On the contrary, the knockdown of MFGE8-L provoked cell migration and invasion (Supplementary Figure 4C). Strikingly, while MDA-MB-231 or HCC1937 expressing MFGE8-L showed a remarkable decrease in migratory and invasive ability compared to control vector, MFGE8-S not only totally abolished the inhibitory function of MFGE8-L but elicited modest pro-migration/ invasion actions on breast cancer cells (**Figure 5A** [↗](#) and Supplementary Figure 4D-4E). We also performed Fluorescent Gelatin Degradation Assays for investigating invadopodia formation and found MFGE8-L up-regulation suppressed breast cancer cells invasion through a layer of extracellular matrix, whereas breast cancer cells with ectopic expression of MFGE8-S acquired enhanced ability to degrade matrix and invasion (**Figure 5B** [↗](#)). Immunofluorescence assays revealed the MFGE8-L forms foci in cytoplasm, particularly enriched near cell-cell contact sites, whereas MFGE8-S is more evenly distributed in the nucleus and cytoplasm (**Figure 5C** [↗](#)). As shown in diagram of domain configuration for two variants, in comparison with MFGE8-L isoform containing N-terminal EGF-like domain and C-terminal twice-repeated F5/8 type C domains, the protein encoded by MFGE8-S loses the second F5/8 type C domain, implying a shortage of the function corresponding to membrane-binding (Supplementary Figure 4F). The different structural bases of two isoforms may also be linked to their distinct cellular function in tumor invasion (Supplementary Figure 4G). Furthermore, we compared transcriptome profiles between MFGE8-L and MFGE8-S ectopically expressing breast cells and found a set of functional genes were down-regulated by MFGE8-L, whereas this negative regulation was abrogated in response to MFGE8-S (Supplementary Figure 4H). The differentially expressed genes between two splice variants were predominantly enriched in cell adhesion molecules and JAK-STAT cascade as revealed by KEGG pathway and Gene ontology analysis (**Figure 5D** [↗](#) and **Figure 5E** [↗](#)). Subsequent experimental validation demonstrated that the activation of STAT1 was hindered by MFGE8-L, but not affected by MFGE8-S (**Figure 5F** [↗](#)). Considering that STAT1 signaling activation plays an important role in tumor cell metastasis via cell adhesion molecules induction (Marimuthu, Lakshmanan et al., 2022) (Chen, Hao et al., 2019), it is highly plausible that MFGE8-L, but not MFGE8-S, impeded breast cancer cell motility and invasive capability in a STAT1-dependent manner.

RBM7 depletion-stimulated aggressiveness of breast cancer is dependent on MFGE8 splicing switch and NF-κB pathway activation

Next, we sought to assess if MFGE8 splicing switch is responsible for acquisition of aggressive feature in RBM7-silenced breast cancer cells. Flag-tagged MFGE8-L or an empty vector was co-expressed in MDA-MB-231 cells together with RBM7 shRNA (Supplementary Figure 5A). Following seeding into transwell chamber, cells transfected with RBM7 shRNA exerted a higher migratory

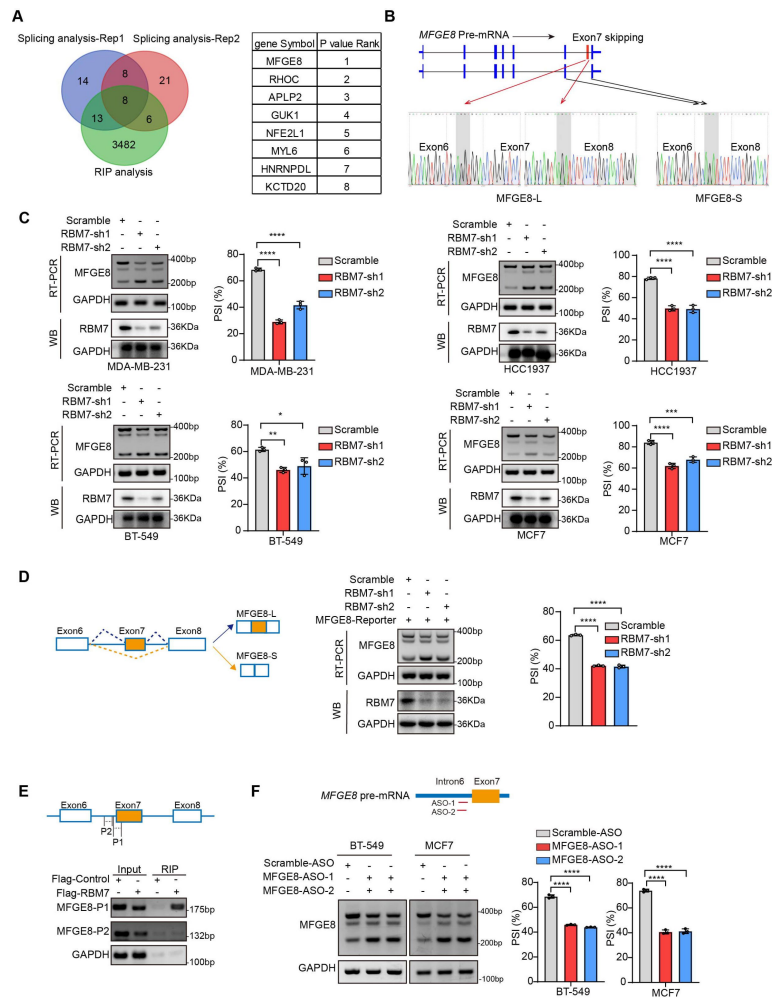


Figure 4.

RBM7 knockdown promoted exon 7 skipping of MFGE8.

(A) Comparison of RBM7-binding candidates from the RNA immunoprecipitation (RIP) sequencing (GSE144075 dataset) and two gene sets containing top 43 differentially AS genes in RNA-seq data of breast cancer cells expressing RBM7 shRNA1/2 as presented by venn diagram. The gene lists on the right were shown according to the FDR P value of top genes with significantly changed AS in RNA-seq. (B) Upper: schematics of human MFGE8 pre-mRNA (NM_005928.4). MFGE8-long isoform included the exon 7 (hereafter referred to as MFGE8-L), whereas it is skipped in MFGE8-short isoform (hereafter referred to as MFGE8-S). The arrow indicates the direction of transcription. Lower: primers were designed on exon 6/8, and RT-PCR was performed to identify PSI changes using RBM7-knockdown or control MDA-MB-231 cells. The base peak diagram of sanger sequencing of RT-PCR results showed the splicing junction sites (filled with gray). (C) Various breast cancer cells lines with stable depletion of RBM7 or control were constructed. Alternative splicing events of MFGE8 regulated by RBM7 were examined by semi-quantitative RT-PCR. The knockdown efficiency of RBM7 expression was detected by Western Blotting. The mean $\pm$ SD of PSIs from three independent repeated experiments were plotted with P value calculated by one-way ANOVA with Dunnett's multiple comparisons test. (D) Left, Schematics of MFGE8 mini-splicing reporter including exons 6-8 and two flanking introns in the 600bp region upstream and downstream of 5'/3' splice sites; middle, RBM7 stable knockdown or control cells were transfected with splicing reporter and collected for RNA extraction after 48h. The splicing changes of exogenous MFGE8 were detected by RT-PCR with specific primers on the reporter gene. The level of RBM7 protein was detected in MCF7 cells by western blotting. P value was calculated by one-way ANOVA with Dunnett's multiple comparisons test from three repeated experiments. (E) As shown in schematic diagram, orange box indicates cassette exon 7 and primers were designed in the putative binding regions P1 and P2. Binding of MFGE8 pre-mRNA with RBM7 was examined by RIP assay in HEK293T cells expressing Flag-RBM7. (F) Upper: the red line in diagram indicates ASOs targeting region which contain UUUUUU residues; down: MCF7 and BT-549 cells were transfected with ASOs targeting MFGE8 pre-mRNA for 48h and then applied for RT-PCR identification. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

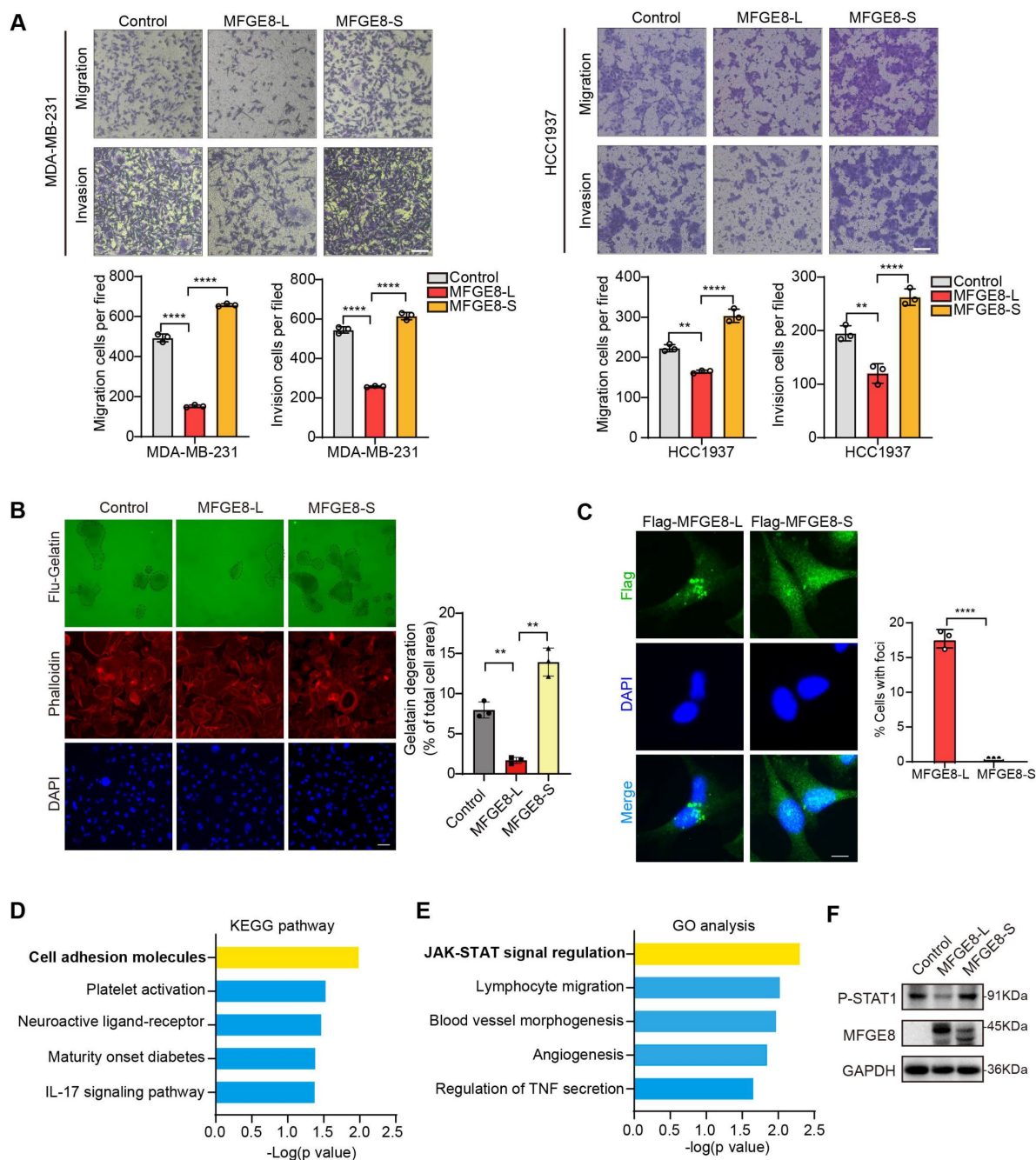


Figure 5.

MFGE8-L inhibited breast cancer cell migration and invasion.

(A) The function of MFGE8-L/S on breast cancer cell migration and invasion were evaluated by transwell assays. (B) Gelatin degradation assay was performed to test the effect of RBM7 knockdown on invadopodia function. 10000 cells were plated onto FITC-gelatin substrates (Green) and cultured for 48 h. Representative images are shown (red, Cy3-phalloidin; blue, DAPI) and the degraded areas were quantified by Image J software. Scale bars = 50 μ m. *P* values were determined by one-way ANOVA with Tukey's multiple comparisons test ($n = 3$). (C) The subcellular localizations of MFGE8 isoforms. MDA-MB-231 cells were transfected with Flag-tagged MFGE8-L or MFGE8-S, and visualized with immunofluorescence assay. Scale bar = 10 μ m. The percent of cells with foci were quantified and plotted (about 80 cells were captured and quantified in both samples). (D-E) The KEGG pathway and Gene ontology analysis of genes up-regulated by MFGE-S compared with MFGE8-L. (F) The expression levels of p-STAT1(Tyr701) in HCC1937 cells overexpressing MFGE8-L/S was detected by western blotting. *P* values were determined by one-way ANOVA with Tukey's multiple comparisons test. $**P < 0.01$, $****P < 0.0001$.

and invasive capability than scramble control, consistent with above data. Importantly, the restoration of MFGE8-L expression diminished the acquired aggressive properties of MDA-MB-231 cells resulted from RBM7 depletion (**Figure 6A** [↗](#)). Similar results were obtained using RBM7-depleted MCF7 cells with or without MFGE8-L complement (**Figure 6A** [↗](#)). To corroborate the role of MFGE8 splicing shift in RBM7-regulated breast cancer cell invasion, we transfected triple-negative breast cancer cells with ASOs directed against RBM7-binding motif and examined the potential impact on cell aggressiveness. The results showed an obvious increase in exon7-skipped variant as compared to the scramble negative control ASOs (Supplementary Figure 5B). Meanwhile, the migrative and invasive ability of breast cancer cells treated with ASOs was significantly boosted (**Figure 6B** [↗](#)), further suggesting that RBM7-knockdown stimulated aggressiveness of breast cancer at least partially relies on the control of MFGE8 AS. However, a series of shRBM7-upregulated secreted factors that have been reported to drive tumor angiogenesis (Gopinathan, Milagre et al., 2015) (Fousek, Horn et al., 2021) (Feng, Ke et al., 2019), were not impacted by MFGE8-L/S (**Figure 6C** [↗](#) and Supplementary Figure 5C). Condition medium derived from breast cancer cells overexpressing MFGE8-L/S showed negligible effect on HUVEC tube formation (Supplementary Figure 5D, implying that there exists additional mechanism underlying breast cancer metastasis attributed to RBM7 reduction.

Given that consecutive gene set enrichment analysis (GSEA) of highest-ranking gene sets based on transcriptomic profile of mRNA expression showed a signature for NF- κ B pathway that was augmented upon depletion of RBM7 (**Figure 6D** [↗](#)), we suspected that the transcriptional activation of downstream genes in NF- κ B signaling known to facilitate tumor angiogenesis might be involved in shRBM7-exerted pro-metastatic function. As expected, the phosphorylation of p65, a core transcription factor well-known to switch on NF- κ B pathway, was notably enhanced by RBM7 knockdown. Meanwhile, I κ B α phosphorylation that can blockade p65 activation was curbed upon RBM7 knockdown (**Figure 6E** [↗](#)). Conversely, RBM7 overexpression impeded the activation of p65 (Supplementary Figure 5E). Yet MFGE8-L/S had no effect on phosphorylation of p65 or I κ B α (Supplementary Figure 5F). We found RBM7 depletion remarkably promoted the expression of IL-1 β as judged by qPCR and ELISA assays (Supplementary Figure 5G-5I). IL-1 β , commonly known as a pro-inflammatory cytokine, could bind to IL-1R and initiate a multistage enzymatic reaction that triggers the activation of NF- κ B pathway (Weber, Kracht et al., 2010) (Guo, Jin et al., 2024). Thus, it is plausible the upregulation of IL-1 β might be a causal factor in RBM7-depletion-induced activation of NF- κ B signaling. We further carried out Matrigel tube formation assays and found NF- κ B inhibitor PDTC drastically reversed the promotion of HUVEC angiogenesis by conditional medium harvested from RBM7-depleted cells (**Figure 6F** [↗](#)-**6G** [↗](#)), implying that RBM7-p65 axis may participate in HUVEC angiogenesis through reprogramming the secretome of breast cancer cells. Similar effect was observed with respect to cell motility and invasion (Supplementary Figure 5J). Taken together, these data above indicated that both MFGE8 splice switch and NF- κ B pathway activation are required for mediating the pro-metastasis activity of RBM7 reduction in breast cancer.

The PSI of MFGE8 exon 7 is decreased in breast cancer patients and positively correlated with RBM7 expression

To study the clinical relevance of MFGE8 isoform switch, we analyzed the PSI changes of cassette exon7 in TCGA breast cancer cohort and found exon7 skipping occurs more frequently in tumor than in normal tissues (**Figure 7A** [↗](#)). Consistently, in comparison with normal counterparts, the splicing of MFGE8 is shifted towards a reduced proportion of MFGE8-L across multiple cancer types, including LUAD, HNSC, COAD, etc (Supplementary Figure 6A). RT-PCR assay was performed in surgical tumor specimens in breast cancer patients and showed increased proportion of exon7 exclusion in most tumor samples when compared to normal tissues (case#1: 86:94; case#2: 84:86; case#3: 79:85; case#4: 63:75; case#5: 69:93; case#6: 71:80) (**Figure 7B** [↗](#)). On the other hand, the lymph node metastases contain a higher proportion of MFGE8 variant with skipped exon7 in comparison with paired primary tumor tissues (case#1: 80:95; case#2: 86:97; case#3: 84:90; case#4:

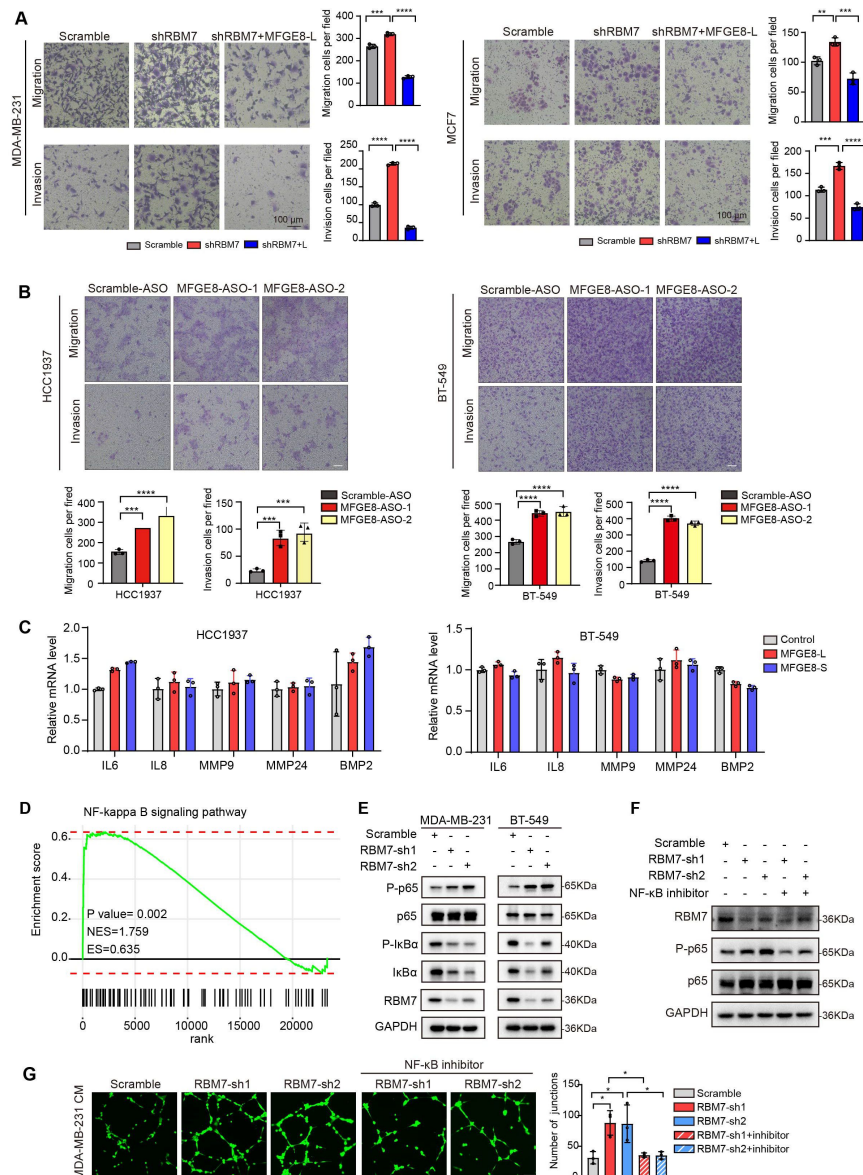


Figure 6.

RBM7 knockdown enhanced aggressiveness of breast cancer relying on MFGE8 splicing switch to the short variant and NF-κB pathway activation. (A) Breast cancer cells MDA-MB-231 and MCF7 were cotransfected with shRNA targeting RBM7 or control and shRNA-resistant Flag-MFGE8-L vectors and then subjected to transwell migration assay or invasion assay. Migrated or invaded cells were counted from random sites of the transwell with Image J. *P* values were determined by one-way ANOVA with Tukey's multiple comparison test. (B) Representative transwell analysis of migrative/invasive capability of breast cancer cells transfected with 500 nM ASO directed against RBM7-binding region in MFGE8 pre-mRNA. Cells migrating through transwell membrane with matrigel or not were imaged and representative images are shown (upper) along with quantification (down). *P* values were determined by one-way ANOVA with Tukey's multiple comparison test. (C) The expression of metastasis-related factors in HCC1937 and BT-549 cells stably expressing MFGE8-L, MFGE8-S or control vector was measured by real-time PCR assay. (D) NF-κB signaling pathway was enrichment via KEGG analysis in RBM7 knockdown breast cancer cells. (E) Western blotting showed the expression of p65, p-p65, IκBα and p-IκBα in NF-κB pathway in RBM7-depleted or control MDA-MB-231 and BT-549 breast cancer cells. (F) Western blotting was performed to test the expression of RBM7, p65 and p-p65 in RBM7-depleted MDA-MB-231 cells treated with or without NF-κB inhibitor PDTC 10 μm for 48h. (G) Representative images of the tube formation of HUVEC cells treated with conditional medium from RBM7-depleted MDA-MB-231 in the presence or absence of 10 μm NF-κB inhibitor PDTC. Scale bars: 100 μm. Quantification of junctions of endothelial network was conducted by ImageJ software. *P* values were determined by one-way ANOVA with Tukey's multiple comparison test (*n* = 3). **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

70:78; case#5: 83:89) (**Figure 7C** [↗](#)). This is coherent with decreased RBM7 expression levels found in breast cancer with lymph node metastasis. In addition, patients with high expression of MFGE8 exon7 had a visibly better overall survival than patients with low expression (**Figure 7D** [↗](#)). We performed Pearson correlation test and revealed a noticeably positive correlation between the expression of RBM7 and exon7 inclusion of MFGE8 in breast cancer (**Figure 7E** [↗](#)), further supporting MFGE8 isoform switch regulated by RBM7.

Discussion

Previous reports have recognized the RNA-binding protein RBM7 as a component of the NEXT complex that could facilitate RNA exosome targeting to a subset of non-coding RNAs and aberrant transcripts (Meola et al., 2016 [↗](#)) (Pascal Preker, 2008), including short-lived exosome substrates and pre-mRNAs, for surveillance and turnover. In addition to this specific characteristic, RBM7 was shown to interact with subunits of 7SK snRNP complex and thus activate P-TEF downstream genes in response to DNA damage (Bugai, Quaresma et al., 2019). Herein, we discover a novel role of RBM7 in alternative splicing control through directly binding to its cognate cis-element in pre-mRNA. Using whole-transcriptome sequencing, we found the depletion of RBM7 contributed to exon exclusion in approximately 70% SE/A5E events, implying RBM7 mainly functions as a splicing activator in the context of SE/A5E in breast cancer. However, the increase of PSI also occurred due to RBM7 knockdown in a smaller fraction of splicing events, such as ZMYND8. We speculated this converse impact may be associated with the intrinsically sophisticated splicing processes driven by distinct core splicing factors, cognate cis-acting element and co-factors in a context-dependent manner (e.g., pre-mRNA structure, base modification, antagonistic or cooperative effect of multiple splicing factors). Similar to many splicing factors, like SR proteins and hnRNPs, RBM7 has an RRM-domain required for target RNA recognition in the N-terminus.

We performed co-IP experiment and found that RBM7 could interact with RNA splicing factor SF3B2, a component of spliceosomal U2 SnRNP complex (Supplementary Figure 6B). Consistent with the AS regulation of MFGE8 by RBM7, the depletion of SF3B2 also promoted exon7 skipping (Supplementary Figure 6C-6D), implying a cooperative effect of the two proteins in regulating MFGE8 splicing. This is in concert with a previous study that RRM domain of RBM7 could bind a proline-rich segment within SF3B2 (Falk, Finogenova et al., 2016). The interaction mode with strong similarity to RBM7^{RRM}-ZCCHC8^{Proline} interaction in the NEXT complex indicated mutually exclusive binding of SF3B2 and ZCCHC8 to RBM7. Thus, RBM7 appears to play dual, but not conflicting, roles during RNA processes depending on its interaction with the spliceosome or exosome.

Currently, the pathobiological function of RBM7 in breast cancer is underexplored except one study that claimed an oncogenic role of RBM7 since its knockdown inhibited breast cancer cell proliferation (Xi et al., 2020 [↗](#)). Nevertheless, in our study, RBM7 knockdown in breast cancer displayed visibly pro-metastasis function with activated NF-κB signaling. More importantly, according to TCGA database (total cases > 1000 used for analysis), RBM7 shows decreased expression in breast cancer, particularly in the metastasis loci and has positive correlation with patients' survival outcomes. It is therefore highly doubtful that RBM7 could serve as an oncogene in breast cancer as previously reported. Moreover, we can't exclude additional biological function of RBM7 in breast cancer. RBM7-targeted silencing for breast cancer would likely be detrimental for neighbouring healthy tissue and immune environment as a result of NF-κB-dependent pro-inflammatory mediators and cytokines secretion. Other splicing events switch triggered by RBM7 might also be linked to tumor development. For example, we found RBM7 regulated alternative splicing of AKT2, which is involved in glucose metabolism (Hay, 2011 [↗](#)). Further investigation is warranted to elaborate more splicing mechanisms of RBM7 in breast cancer.

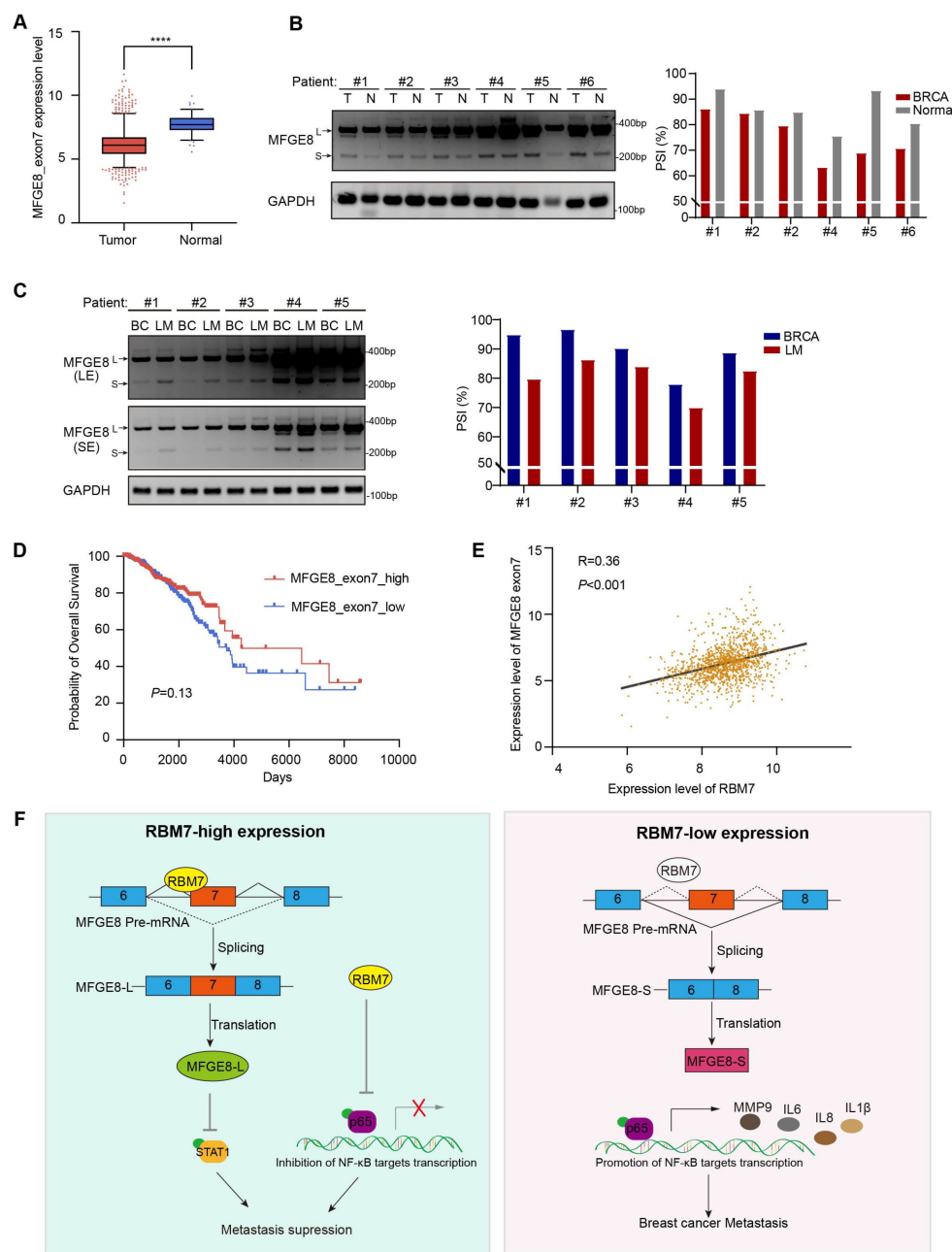


Figure 7.

Splicing shift of MFGE8 toward exon7 exclusion was negatively correlated with RBM7 expression in patients with breast cancer.

(A) MFGE8 exon7 expression level was analyzed in breast carcinoma in comparison with normal tissues based on the TCGA dataset. Plotted are the mean $\pm$ SD from 114 normal tissues and 1094 tumor tissues, with **** $P < 0.0001$ as determined by unpaired Student's *t* test. (B) The splicing alteration of MFGE8 in 6 pairs of primary breast cancer tissues and adjacent normal tissues was examined using RT-PCR. The quantification of PSI values was based on relative band intensities using ImageJ software. (C) The splicing alteration of MFGE8 in primary breast cancer tissues and corresponding lymph node metastases was identified by RT-PCR assays. The quantification of PSI values was determined by ImageJ software. (D) The correlation between the expression of MFGE8 exon7 and overall survival (OS) of breast cancer patients was analyzed based on the TCGA dataset. P value was assessed with Mantel-Cox log-rank test. (E) Correlation of RBM7 expression with MFGE8 exon7 levels was analyzed using Pearson's correlation coefficient method. (F) The mechanistic model of how RBM7 regulated metastasis of breast cancer through regulating MFGE8 splicing switch.

In conclusion, we discovered that down-regulated RBM7 and its cognate elevated exon7-exclusion MFGE8 were tightly related to enhanced breast cancer metastatic potential and poor prognosis of patients. MFGE8-L, the canonical variant, possessed the capability to trigger STAT1 phosphorylation and inhibit the motility and invasion of breast cancer cells, whereas splice switch to MFGE8-S was found loss-of-function for this tumor-suppressive role. At another layer, RBM7 reduction led to NF- κ B pathway activation, resulting in enhancement of breast tumor aggressiveness. Overall, our findings implicated that activated NF- κ B pathway worked in concert with MFGE8 splicing transition to spur metastasis in RBM7-depleted breast cancer. Additionally, although the detailed clinical significance underlying their cooperation need further in-depth investigation, we propose that RBM7 and its regulated splicing variants might serve as potential therapeutic biomarker and prognostic indicator in breast cancer patients.

Materials and Methods

Cell culture

Human breast cancer cell lines (MDA-MB-231, MCF7, HCC1937 and BT-549) and human embryonic kidney cell HEK293T in this article were obtained from the American Type Culture Collection (ATCC). MDA-MB-231 cells were cultured without carbon dioxide at 37°C in Leibovitz's L-15 medium (Gibco) contained 10% FBS (SINSAGE TECH). MCF7 and HEK293T cells were cultured with DMEM medium (Gibco) supplemented with 10% FBS. HCC1937 cells were cultured with RPMI-1640 media (Gibco) supplemented with 10% FBS, while BT-549 was maintained with RPMI-1640 media (Gibco) supplemented with 10% FBS and 0.023 U/ml insulin. HUVECs were cultured with Endothelial Cell Medium (ScienCell) supplemented with ECGS (ScienCell), P/S solution (ScienCell) and 10% FBS (ScienCell). Except for MDA-MB-231, all the cells above were cultured in a 37 °C humidified incubator.

Stable breast cancer cell lines construction

To generate RBM7 overexpression vector, its coding regions (NM_016090.4) were cloned into pCDH-CMV-MCS-EF1-Puro with N-terminal Flag tag or HA-tag. To construct MFGE8-L/S vectors, the coding regions of MFGE8-L (NM_005928.4) and MFGE8-S (NM_001114614.3) were cloned into pCDH-CMV-MCS-EF1-Puro with N-terminal Flag tag. For RBM7 knockdown, target-specific shRNA (RBM7-sh1: 5'-CGAAAGGACTATGGATAACAT-3'; RBM7-sh2: 5'-CTACCTAGCAGATAGACA TTA-3') was cloned into pLKO.1 vector. And HEK293T cells were transfected these over-expression (or low-expression) lentivirus plasmid together with lentivirus packaging plasmids pPAX2 and pMD2 according to the manufacturer's instructions. Viruses were collected 48 hours after transfection and then to infect breast cancer cell line with 8 ug/ml polybrene (beyotime biotechnology) for 24 hours. Positive cells were selected with 3 ug/ml puromycin.

Western Blot assay

Cells were lysed with RIPA buffer (10 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.1% SDS, 1% Triton X-100, and 1% sodium deoxycholate) supplemented with protein tyrosine phosphatase inhibitor Na3VO4 (1mM), serine protease inhibitor PMSF (1mM) and protease inhibitor Cocktail. Immunoblotting assay was performed using standard methods. The following antibodies were used: anti-RBM7 (Sigma-Aldrich, HPA013993), anti-Flag (Sigma-Aldrich, F1804), anti-GAPDH (Proteintech, 10494-1-AP), anti- α -Tubulin (Proteintech, 11224-1-AP), anti-HA tag (Abcam, ab91110), anti-Phospho-NF- κ B p65 (Cell Signaling Technology, #3033), anti-NF- κ B p65 (Cell Signaling Technology, #8242), anti-I κ B α (Cell Signaling Technology, #4814), anti-Phospho-I κ B α (Ser32) (Cell Signaling Technology, #2859), anti-Phospho-stat1 (Tyr701) (Cell Signaling Technology, #8062), and anti-MFGE8 (R&D systems, MAB2767). The protein bands were detected with ECL enhanced chemiluminescence reagent kits (NCM, Biotech), visualized using the MiniChemiluminescent image system (Beijing Sage Creation, China).

RNA isolation, RNA sequencing, and PCR

Total RNA from cells and tumor tissues was extracted using Trizol (Invitrogen) according to the manufacturer's instructions. Total mRNA was cleaned up using RNeasy Lipid Tissue Mini Kit (Qiagen, USA). And RNA-seq was carried out on platform of Novogene Illumina sequencing. cDNAs were obtained using *Evo M-MLV* Plus 1st Strand cDNA Synthesis Kit (Accurate Biotechnology, AG11615) with random primers. qRT-PCR was carried out using MonAmpTM Universal ChemoHS Specificity Plus qPCR Mix. GAPDH mRNA was examined as control for normalization. Gene expression relative to GAPDH was calculated using the formula $2^{-\Delta\Delta CT}$. For identified alternative splicing events, primers of the upstream and downstream of variable exons or introns were designed and PCR assay was performed. PCR products were subject to DNA agarose gel electrophoresis. AS events change of percent-spliced-in (PSI) values were the ratio of retained splices to total transcripts. The amount of each splicing isoform was measured with ImageJ. All the primers are showed in Table1.

Migration and invasion assays

For transwell migration assay, 10×10^4 MDA-MB-231 or other breast cancer cells (MCF7, BT-549 and HCC1937) were plated in the upper chamber of 6.5 mm insert (Costar, USA) with 8.0 μ m polycarbonate membrane with 100 μ L serum-free medium. For invasion assay, 10×10^4 breast cancer cells were seeded in the top chamber of each Matrigel-coated insert (BD Biosciences, USA). After 24 hours or 48 hours of incubation at 37 °C, migrated or invaded cells were fixed with 4% PFA and stained in 0.1% crystal violet. The cell number was counted by ImageJ software in 3 views from 3 independent experiments.

Wound healing

Knockdown RBM7 in MDA-MB-231 and 4T1 cells or control cells were implanted into 6-well culture dishes. When the cells grew about 90% confluence, lines of the same width were drawn on the bottom of the culture dish using autoclaved sterile tips. Images were captured at 0 and 36 hours after the wounding. Data shown were representative of three independent repeats.

Gelatin Degradation Assay

To evaluate the ability of breast cancer cells to form invadopodia and degrade matrix, QCM Gelatin Invadopodia Assay (ECM670-1, Merck Millipore) was performed according to the manufacturers' protocol. Briefly, 1×10^4 MFGE8-L/S-overexpressing or control MDA-MB-231 cells were suspended in L-15 medium with 10% FBS and seeded into the pre-treated 96-well plate. After 48 hours, the cells were washed twice with PBS. Fix cells with 3.7% formaldehyde for 30 minutes. Then these cells were stained respectively with TRITC-phalloidin and DAPI for 1 hour. The ability of cells to degrade the matrix was imaged using immunofluorescence microscopy (LEICA DMi8, Germany) and the area of invadopodia was measured using the Image J software.

Tube formation of HUVECs assay

Capillary tube formation assay was used to observe angiogenesis of HUVECs. HUVECs were seeded into 48-well plate coated with growth factor reduced Matrigel (BD Biosciences). HUVECs cells were treated with conditioned medium for 4-5 hours and then formation of capillary-like tubes was observed and stained with calcein for 30 minutes. Images of blood vessels were taken using fluorescence microscopy. And the tube length was analyzed by ImageJ software.

Immunofluorescence assay

After seeded on coverslips for 36 hours, the MDA-MB-231 cells were immersion fixed with 4% PFA for 20 minutes at room temperature, washed 3 times in PBS. Then the cells were treated with 0.2 % Triton-X100 in PBS for 10 minutes and blocked with 3% BSA for 10 minutes. Next the cells were

immersed in primary antibody at 4°C overnight, washed in PBS 3 times for 5 minutes each, and incubated with secondary antibody labelled with Alexa fluorophore 488/592 for 1 hour at RT. After washing 3 times in PBS, the cells were dyed with DAPI. In addition, antibodies targeting RBM7 (Sigma, HPA013993), and Flag (Sigma, 3165) were used in the assay. The pictures are captured with fluorescent microscope (Leica, USA).

RNA immunoprecipitation assay

RNA immunoprecipitation was performed as previously described. In brief, HEK293T cells were transfected Flag-RBM7 or control plasmid. After 48 hours of transfection, Flag-RBM7 or control cells were collected and cross-linked by 1% formaldehyde, homogenized in lysis buffer (50 mM Tris-HCl Ph 7.5, 0.4 M NaCl, 1 mM EDTA, 1mM DTT, 0.5% TritonX-100, 10% Glycerol supplemented with RNase inhibitor and broad-spectrum protease inhibitors Cocktail), and broke with an ultrasonic crusher. And solubilized cell lysate was obtained and precleared by ProteinG-agarose beads together with nonspecific tRNA to remove nonspecific binding. Then binding RNAs with RBM7 protein were precipitated with Anti-Flag M2 Affinity beads. Co-precipitated RNAs were subject to RT-PCR. Input controls (total RNAs) and Flag-controls were assayed simultaneously to prove that the detected signals were the result of RNAs specifically binding to RBM7. The gene-specific primers were designed MFGE8-RIP1-F:5'-tatttatgccccctcaccgc-3'; MFGE8-RIP1-R:5'-atgttggtgctgctgtaat-3';MFGE8-RIP2-F:5'-ctgaggctcaccgaagtag-3'; MFGE8-RIP2-R:5'-GTAGGATGCCACAACTG-3'). GAPDH mRNA was as negative control (GAPDH-F 5'-GGGGAAGGTGAAGGTCG-G-3'; GAPDH-R 5'-TTGAGGTCAATGAAGGG-3').

Antisense oligonucleotide (ASO)-mediated splicing blockade

MFGE8-ASOs were synthesized with a full-length phosphorothioate backbone and 2'-O-methyl modified ribose molecules (GenScriptBiotech Corp). ASO was transfected at final concentration of 500 nM using Lipofectamine3000 (Thermo Fisher Scientific) according to the manufacturer's instructions. The MCF7 cells were harvested for RNA and splicing assays 48 h after transfection.

Animal experiments

All procedures and experimental protocols involving mice were approved by the Institutional Animal Care and Use Committee (IACUC) of Dalian Medical University. Female BALB/C mice (6 weeks old, 4-5 mice per group as indicated in figures and/or figure legends) were injected into tail vein with murine 4T1 breast cancer cells (6×10^5 cells) to develop mice model of pulmonary metastasis. After about 3 weeks, all mice were euthanized, and lungs were harvested, fixed and paraffin embedded. After lungs were H&E stained, the number of metastatic lesions were enumerated by pathologist using brightfield microscopy.

Clinical tissues samples collection

All human breast cancer tissues or normal tissue were obtained with written informed consent from patients or their guardians prior to participation in the study. The Institutional Review Board of the Dalian Medical University approved use of the tumor specimens in this study.

Bioinformatics analysis

RBM7 expression data and survival data (including OS and DFS) were obtained from the METABRIC database using the Cbioportal web tool. Based on the expression level of RBM7, the patients were divided into four categories (i.e., Q1, Q2, Q3, Q4) by quartiles, and then log-rank was employed to compare the survival differences of patients in different categories. Cbioportal cited (Cerami E et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* 2012 May;2(5):401-4). Through the analysis of alternative splicing events in RBM7 knocked-down MDA-MB-231 cells compared to control, exon skipping events were screened out, and then selected the sequence of the 500nt region upstream of the

alternative exon for subsequent base sequence enrichment analysis. The XSTREME tool provided by the online tool meme-suite (<https://meme-suite.org/meme/>) was used to perform Motif Discovery and Enrichment Analysis. E-value < 0.05 as the significance standard to output the results.

Statistical analyses

Statistical analyses and statistical graphs were conducted by GraphPad Prism 8 software. To compare the differences between the two variables, Student's T-test, one-way ANOVA with Dunnett multiple comparisons, and one-way ANOVA with Tukey's multiple comparisons according to the specific situation were applied. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$ indicated the significant difference.

Data availability

The authors declare that all the data supporting the findings of this study are available within the article and its supplementary information files. RNA-seq data in this study have been deposited in Gene Expression Omnibus of NCBI with the accession code GSE248938.

Author contributions

YW and YQ conceived the project and designed the experiments. FH, JY, ZD, KW, and DC performed most of the experiments, whereas CC, JZhang, JZhao and WZ performed data analysis. YW and YQ wrote the manuscript.

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Study Approval

The Institutional Animal Care and Use Committee of the Dalian Medical University approved use of animal models in this study. All human tumor tissues were obtained with written informed consent from patients or their guardians prior to participation in the study. The Institutional Review Board of the Dalian Medical University approved use of the tumor specimens in this study.

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Editors

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Reviewer #1 (Public Review):

Fang Huang et al found that RBM7 deficiency promotes metastasis by coordinating MFG-E8 splicing switch and NF- κ B pathway in breast cancer by utilizing clinical samples as well as cell and tail vein injection models.

This study uncovers a previously uncharacterized role of MFG-E8 splicing alteration in breast cancer metastasis, and provides evidence supporting RBM7 function in splicing regulation. These findings facilitate the mechanistic understanding of how splicing dysregulation contributes to metastasis in cancer, a direction that has increasingly drawn attention recently, and provides a potentially new prognostic and therapeutic target for breast cancer.

<https://doi.org/10.7554/eLife.95318.2.sa1>

Reviewer #2 (Public Review):

In this manuscript, the authors reported the biological role of RBM7 deficiency in promoting metastasis of breast cancer. They further used a combination of genomic and molecular biology approaches to discover a novel role of RBM7 in controlling alternative splicing of many genes in cell migration and invasion, which is responsible for the RBM7 activity in suppressing metastasis. They conducted an in-depth mechanistic study on one of the main targets of RBM7, MFGE8, and established a regulatory pathway between RBM7, MFGE8-L/MFGE8-S splicing switch, and NF- κ B signaling cascade. This link between RBM7 and cancer pathology was further supported by analysis of clinical data.

Overall, this is a very comprehensive study with lots of data, and the evidence is consistent and convincing. Their main conclusion was supported by many lines of evidence, and the results in animal models are pretty impressive.

<https://doi.org/10.7554/eLife.95318.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Reviewers' Comments:**Reviewer #1 (Remarks to the Author):***Summary:*

Fang Huang et al found that RBM7 deficiency promotes metastasis by coordinating MFGE8 splicing switch and NF- κ B pathway in breast cancer by utilizing clinical samples as well as cell and tail vein injection models.

Strengths:

This study uncovers a previously uncharacterized role of MFGE8 splicing alteration in breast cancer metastasis, and provides evidence supporting RBM7 function in splicing regulation. These findings facilitate the mechanistic understanding of how splicing dysregulation contributes to metastasis in cancer, a direction that has increasingly drawn attention recently, and provides a potentially new prognostic and therapeutic target for breast cancer.

We thank the reviewer for appreciating the novelty and importance of this study, and have provided new data to address the following concerns raised by the reviewer.

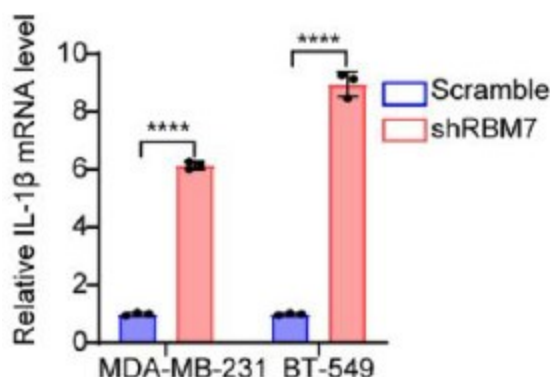
Weaknesses:

This study can be strengthened in several aspects by additional experiments or at least by further discussions. First, how RBM7 regulates NF- κ B, and how it coordinates splicing and canonical function as a component of NEXT complex should be clarified. Second, although the roles of MFGE8 splicing isoforms in cell migration and invasion have been demonstrated in transwell and wound healing assays, it would be more convincing to explore their roles in vivo such as the tail vein injection model. Third, the clinical significance would be considerably improved, if the therapeutic value of targeting MFGE8 splicing could be demonstrated.

We're thankful for the constructive suggestions. A preliminary study on the mechanism by which RBM7 regulates NF- κ B pathway is already underway. We found RBM7 depletion remarkably promoted the expression of IL-1 β as judged by qPCR and ELISA assays (new Figure S5G- S5I, also see below). IL-1 β , commonly known as a pro-inflammatory cytokine, could bind to IL-1R and initiate a multistage enzymatic reaction that triggers the activation of NF- κ B pathway (Axel Weber, 2010) (Qing Guo, 2024). Thus we speculated that the upregulation of IL-1 β might be a causal factor in RBM7-depletion-induced activation of NF- κ B signaling. It will be interesting to determine the complete molecular mechanism in our future study. In addition, we performed a co-IP experiment and found that RBM7 could interact with RNA splicing factor SF3B2, a component of spliceosomal U2 snRNP complex (new Figure S6B, also see below). Consistent with the AS regulation of MFGE8 by RBM7, the depletion of SF3B2 also promoted exon7 skipping, implying a cooperative effect of the two proteins in regulating MFGE8 splicing (new Figure S6C-6D, also see below). This is in concert with a previous study that RRM domain of RBM7 could bind a proline-rich segment within SF3B2 (Falk, Finogenova et al., 2016). The interaction mode with strong similarity to RBM7RRM-ZCCHC8Proline interaction in the NEXT complex indicated mutually exclusive binding of SF3B2 and ZCCHC8 to RBM7. Thus, RBM7 appears to play dual, but not conflicting, roles during RNA processes depending on its interaction with the spliceosome or exosome (see line 427-437 in the new manuscript).

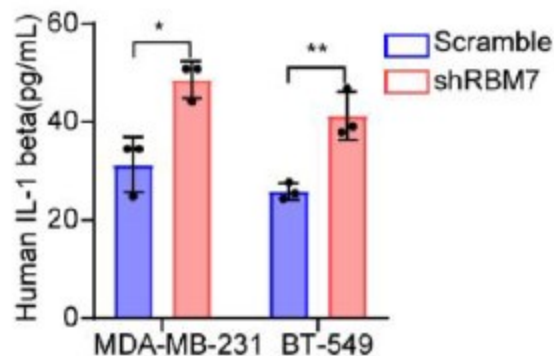
Author response image 1.

The mRNA levels of IL-1 β in MDA-MB-231 or BT549 cells with stable RBM7 knockdown or control vector were examined by qRT-PCR approach.



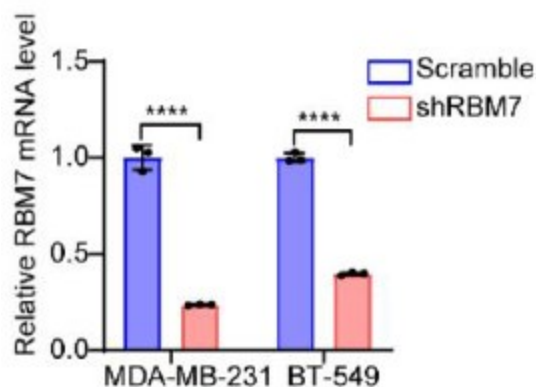
Author response image 2.

Supernatants from RBM7-knockdown MDA-MB-231 or BT549 cells were collected and protein expression of IL-1 β was measured by ELISA kit.



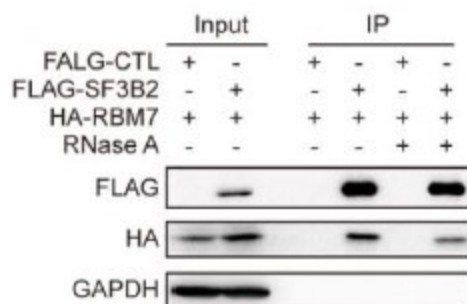
Author response image 3.

The knockdown efficiency of RBM7 in two breast cancer cell lines were determined by qRT-PCR approach.



Author response image 4.

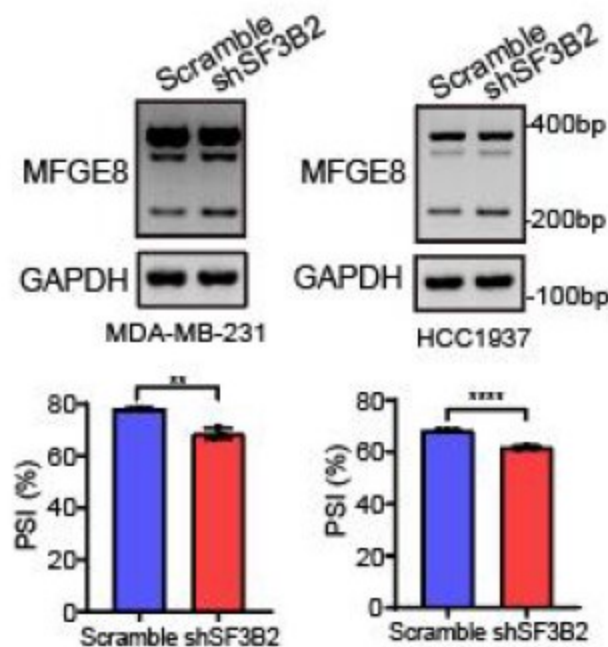
Immunoprecipitation assay was performed in breast cancer cells expressing HA-RBM7 and Flag-SF3B2 or empty vector. The Flag-tagged precipitated complexes and lysates were analyzed through western blotting.



Author response image 5.

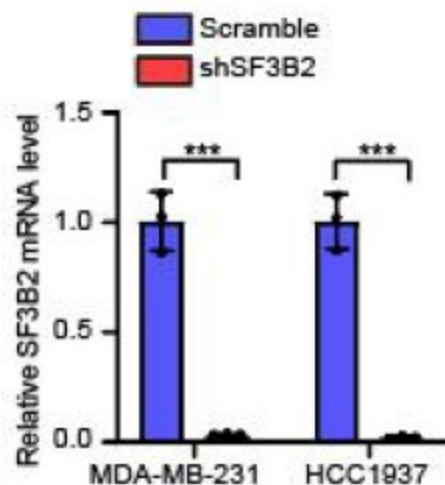
The splicing shift of MFGE8 upon SF3B2 knockdown in breast cancer cells was examined by RT-PCR approach. The mean $\pm$ SD of PSI values derived from three independent replicates is

shown.



Author response image 6.

The SF3B2 knockdown efficiency was examined by qRT-PCR.

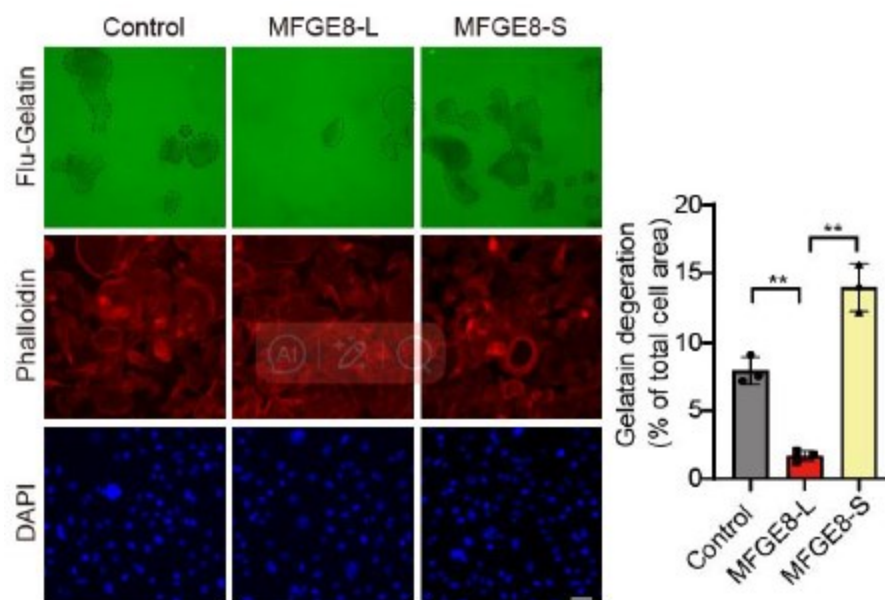


To further corroborate the roles of two MFGE8 isoforms in cell invasion, we have performed Fluorescent Gelatin Degradation Assays for investigating invadopodia formation. Consistent with the transwell assay results, MFGE8-L up-regulation suppressed breast cancer cells invasion through a layer of extracellular matrix, whereas breast cancer cells with ectopic expression of MFGE8-S acquired enhanced ability to degrade matrix and invasion (new Figure 5B, also see below). In addition, to determine the therapeutic value of targeting MFGE8 splicing, we transfected triple-negative breast cancer cells with ASOs targeting RBM7-binding motif and examined the potential impact on cell aggressiveness. The results showed an obvious increase in exon7-skipped variant of MFGE8 as compared to the scramble negative

control ASOs, meanwhile, the migrative and invasive ability of breast cancer cells treated with splice-targeting ASOs was significantly boosted (new Figure 6B and S5B, also see below), further suggesting that RBM7-knockdown stimulated aggressiveness of breast cancer at least partially relies on splicing switch of MFGE8.

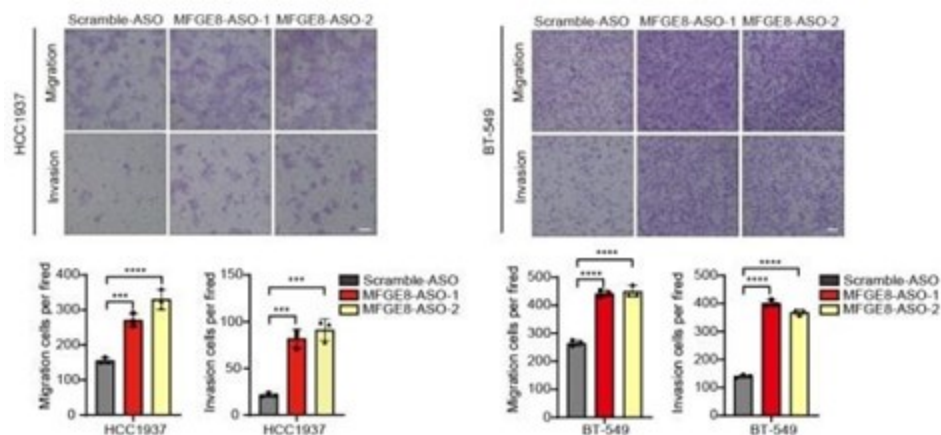
Author response image 7.

Gelatin degradation assay was performed to test the effect of RBM7 knockdown on invadopodia function. 10000 cells were plated onto FITC-gelatin substrates (Green) and cultured for 48 h. Representative images are shown (red, Cy3-phalloidin; blue, DAPI) and the degraded areas were quantified by Image J software. Scar bars= 50 μ m. P values were determined by one-way ANOVA with Tukey's multiple comparison test (n = 3).



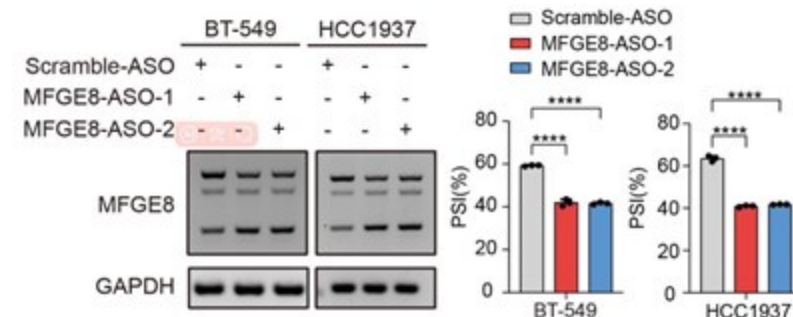
Author response image 8.

Representative transwell analysis of migrative/invasive capability of breast cancer cells transfected with 500 nM ASO directed against RBM7-binding region in MFGE8 pre-mRNA. P values were determined by one-way ANOVA with Tukey's multiple comparison test.



Author response image 9.

RT-PCR quantification of two MFGE8 isoforms after transfecting breast cancer cells with 500 nM ASO directed against RBM7-binding region in MFGE8 pre-mRNA. P values were calculated by one-way ANOVA with Tukey's multiple comparison test.



The minor concerns

(1) Several figure legends do not match with the images, for example, Figure 2K, Figure 4, Figure 7D, and 7E, and the description of Figure 7F is missing in the text.

As suggested by the reviewer, we have checked all of the figure legends carefully and corrected all of the misinterpretation.

(2) The statistical methods for Figure 1A and Figure 1B should be indicated.

As suggested by the reviewer, we have included the statistical methods for Figure 1A and 1B in Figure 1 legend. Data in Figure 1A and 1B are presented as means $\pm$ SD and P values were obtained by Mantel-Cox log-rank test.

(3) The molecular weight of the proteins in the Western Blot images should be marked.

As suggested by the reviewer, we have added the molecular weight of proteins in all of the western blot images.

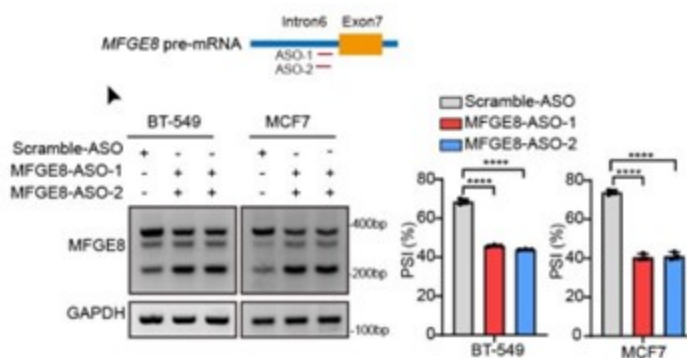
(4) The sequences where RBM7 binds on MFGE8 RNA should be clearly indicated.

We thank the reviewer for this question. We analyzed the sequence of alternative exon 7 and the motifs nearby its 5' or 3' splice sites, and found two RBM7 potentially binding motifs are positioned in proximal to the pseudo 3' splice site. Subsequent RT-PCR for the precipitation in RIP assays confirmed RBM7 could bind to the upstream sequence containing 5'-UUUCUU-3' motifs adjacent to intron6/exon7 junction of MFGE8 cassette exon, but not another region nearby it. To pinpoint the location for the potential cis-element for AS regulation by RBM7, we designed antisense oligonucleotides (ASOs) to block RBM7 potentially binding sites (UUUCUU). As shown in revised Figure 4F, when compared to scramble ASO, targeting ASOs contributed to the exclusion of exon7. Additionally, we constructed an exogenous MFGE8 splicing reporter containing exon 6-8 and partial intron sequences to determine the binding site for AS regulation by RBM7. The depletion of RBM7 still induced the splicing shift of the minigene reporter by elevating MFGE8-S variant. While the binding motif UUUCUU was removed or mutated, RBM7 failed to affect the splicing outcomes of MFGE8 (new Figure S3C, also see below). Due to its close proximity to 3' splice site, UUUCUU residues bound by RBM7

is very likely to participate in spliceosome assembly at the upstream 3' splice site of exon7, which may explain why disruption of the motif led to almost complete exon7 skipping. The above data suggested that RBM7 regulated the exon skipping of MFGE8 by binding to UUUCUU located six nucleotides upstream of the 3' splice-site of exon7.

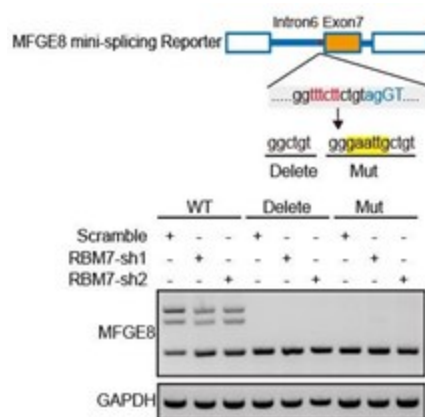
Author response image 10.

Upper: the red line in diagram indicates ASOs targeting region which contains UUUCUU; down: MCF7 and MDA-MB-231 cells were transfected with ASOs targeting MFGE8 pre-mRNA for 48h and then applied for RT-PCR identification. P values were determined by one-way ANOVA with Tukey's multiple comparison test.



Author response image 11.

Upper: MFGE8 min-splicing reporters with mutation in the RBM7 binding site or a non-specific binding were generated and shown in cartoon; down: RT-PCR assays were performed to identify the splicing outcomes of MFGE8 reporter while RBM7 was depleted in breast cancer cells.



(5) Some typos, graphic errors, and sentences are hard to understand and need to be corrected, such as lines 80-81, 249-250, line 221 "motfs", line 319 "RBM4". Please carefully proofread and revise the entire manuscript.

As suggested by the reviewer, we have corrected typos and graphic errors mentioned above. In addition, this manuscript was also extensively edited to improve grammar and sentence structure.

(6) Define the abbreviations when they first appear, such as MFGES-L, RBM, etc.

We thank the reviewer for raising this point. We have defined the abbreviations when firstly presented in the manuscript.

Reviewer #2 (Public Review):

Summary:

In this manuscript, the authors reported the biological role of RBM7 deficiency in promoting metastasis of breast cancer. They further used a combination of genomic and molecular biology approaches to discover a novel role of RBM7 in controlling alternative splicing of many genes in cell migration and invasion, which is responsible for the RBM7 activity in suppressing metastasis. They conducted an in-depth mechanistic study on one of the main targets of RBM7, MFGES, and established a regulatory pathway between RBM7, MFGES-L/MFGES-S splicing switch, and NF-κB signaling cascade. This link between RBM7 and cancer pathology was further supported by analysis of clinical data.

Strengths:

Overall, this is a very comprehensive study with lots of data, and the evidence is consistent and convincing. Their main conclusion was supported by many lines of evidence, and the results in animal models are pretty impressive.

Weaknesses:

However, there are some controls missing, and the data presentation needs to be improved. The writing of the manuscript needs some grammatical improvements because some of the wording might be confusing.

We thank the reviewer for the positive comments on this work, and have addressed all the concerns raised by the reviewer.

Specific comments:

(1) Figure 2. The figure legend is missing for Figure 2C, which caused many mislabels in the rest of the panels. The labels in the main text are correct, but the authors should check the figure legend more carefully. Also in Figure 2C, it is not clear why the authors choose to examine the expression of this subset of genes. The authors only refer to them as "a series of metastasis-related genes", but it is not clear what criteria they used to select these genes for expression analysis.

We thank the reviewer for raising this question. We have included the figure legend for Figure 2C and improved other figure legends throughout the article. For the second question, since gene ontology analysis of RNA-seq data in RBM7-depleted breast cancer cells showed that a series of differentially expressed genes were enriched in metastasis-associated processes, we identified the expression of this subset of genes in breast cancer cells in the presence or absence of RBM7 by heatmap differential analysis based on qRT-PCR results. To clarify this point and address the reviewer's concern, we have improved the relevant description of this part (see line 174-180 in the new manuscript).

(2) Line 218-220. The comparison of PSI changes in different types of AS events is misleading. Because these AS events are regulated in different mechanisms, they cannot draw the conclusion that "the presence of RBM7 may promote the usage of alternative splice sites". For example, the regulators of SE and IR may even be opposite, and thus

they should discuss this in different contexts. If they want to conclude this point, they should specifically discuss the SE and A5SS rather than draw an overall conclusion.

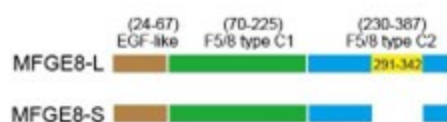
We are thankful for the reviewer's valuable comment. According to the suggestion, we have removed the overall conclusion and corrected to discuss in SE and A5SS.

(3) In the section starting at line 243, they first referred to the gene and isoforms as "EFG-E8" or "EFG-E8-L", but later used "EFGE8" and "EFGE8-L". Please be consistent here. In addition, it will be more informative if the authors add a diagram of the difference between two EFGE8 isoforms in terms of protein structure or domain configuration.

As suggested by the reviewer, we keep using the name "MFGE8-L" for the canonical MFGE8 isoform and "MFGE8-S" for the truncated isoform in this manuscript. In addition, to clarify the structural basis for the different tumor invasion-related functions of two MFGE8 isoforms, we have included a diagram of their domain configuration in new Figure S4F and predicted protein structure in new Figure S4G. The details in the revised manuscript are given below:

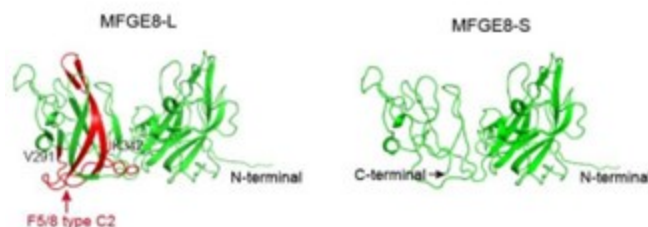
Author response image 12.

Schematic diagram of the domain composition of two MFGE8 isoforms. Upper: the full-length variant with exon7 indicated by yellow square; down: the truncated variant with exon7 skipping.



Author response image 13.

The model structure of two MFGE8 isoforms was implemented using SwissModel software. The F5/8 type C2 protein domain excluded from MFGE8-S variant was marked in red.



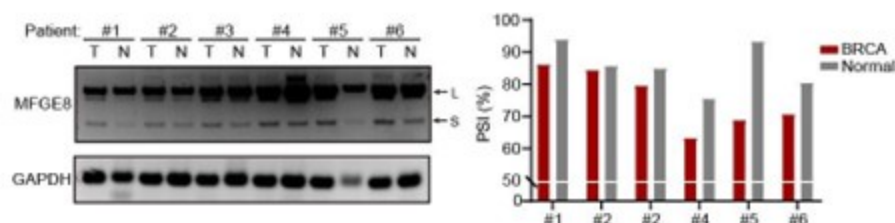
(4) Figure 7B and 7C. The figures need quantification of the inclusion of MFGE exon7 (PSI value) in addition to the RT-PCR gel. The difference seems to be small for some patients.

As suggested by the reviewer, we have included the relative quantification of PSI for endogenous MFGE8 in breast cancer patients and found increased proportion of exon7 exclusion in most tumor samples when compared to normal tissues (case#1: 86:94; case#2: 84:86; case#3: 79:85; case#4: 63:75; case#5: 69:93; case#6: 71:80) (new Figure 7B, also see below). On the other hand, we have expanded the number of metastatic breast cancer cases and quantified the the AS events within MFGE8 by analyzing the PSI values. The lymph node metastases contain a higher proportion of MFGE8 variant with skipped exon7 in comparison

with paired primary tumor tissues (case#1: 80:95; case#2: 86:97; case#3: 84:90; case#4: 70:78; case#5: 83:89) (Figure 7C). This is coherent with decreased RBM7 expression levels found in breast cancer with lymph node metastasis.

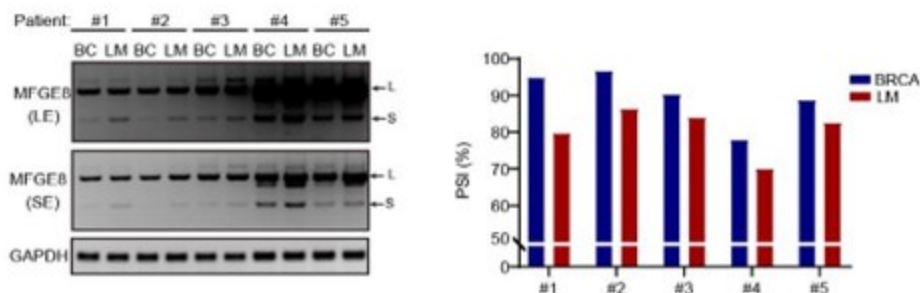
Author response image 14.

The splicing alteration of MFGE8 in 6 pairs of primary breast cancer tissues and adjacent normal tissues was examined using RT-PCR. The quantification of PSI vales was based on relative band intensities using Image J software.



Author response image 15.

The splicing alteration of MFGE8 in primary breast cancer tissues and corresponding lymph node metastases was identified by RT-PCR assays. The quantification of PSI vales wa determined by Image J software.



Minor comments:

The writing in many places is a little odd or somewhat confusing, I am listing some examples, but the authors need to polish the whole manuscript more to improve the writing. 1. Line 169-170, "...followed by profiling high-throughput transcriptome by RNA sequencing", should be "followed by high-throughput transcriptome profiling with RNA sequencing". 2. Line 170, "displayed a wide of RBM7-regulated genes were enriched...", they should add a "that" after the "displayed" as the sentence is very long. 3. Line 213, "PSI (percent splicing inclusion)" is not correct, PSI stands for "percent spliced in". 4. Line 216-217, the sentence is long and fragmented, they should break it into two sentences. 5. Line 224, the "tethering" should be changed to "recognizing". There is a subtle difference in the mechanistic implication between these two words. 6. Line 250, should be changed to "...in the ratio of two MFGE8 isoforms".

We thank the detailed comments from the reviewer. The points mentioned above has been addressed one by one and this manuscript was also extensively edited to improve grammar and sentence structure for better understanding.

References

- Axel Weber PW, Michael Kracht* (2010) Interleukin-1 (IL-1) Pathway. SCIENCE SIGNALING.
- Qing Guo¹, Yizi Jin^{1,2}, Xinyu Chen³, Xiaomin Ye⁴, Xin Shen⁵, Mingxi Lin^{1,2}, Cheng Zeng^{1,2}, Teng Zhou^{1,2} and Jian Zhang^{1,2} (2024) NF- κ B in biology and targeted therapy: new insights and translational implications. *Signal Transduction and Targeted Therapy*.
- Falk S, Finogenova K, Melko M, Benda C, Lykke-Andersen S, Jensen TH, Conti E (2016) Structure of the RBM7–ZCCHC8 core of the NEXT complex reveals connections to splicing factors. *Nature Communications*.

<https://doi.org/10.7554/eLife.95318.2.sa3>